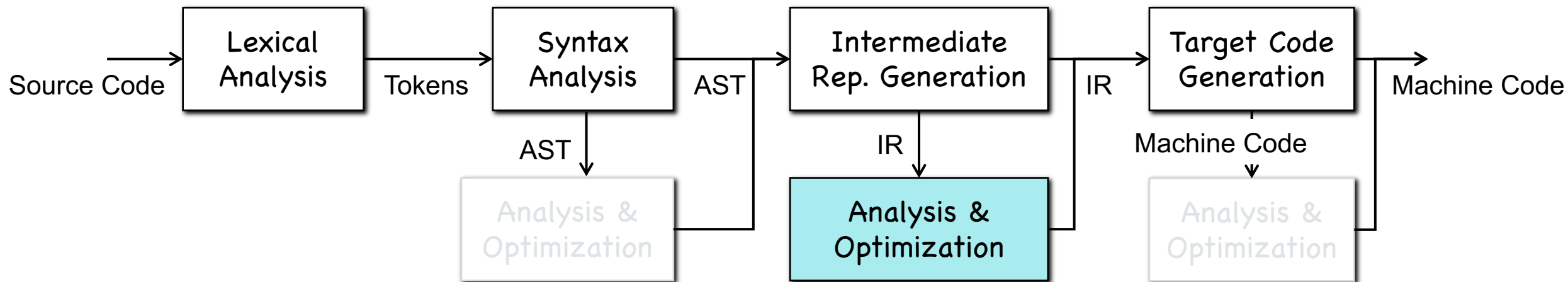


Chapter 9-1

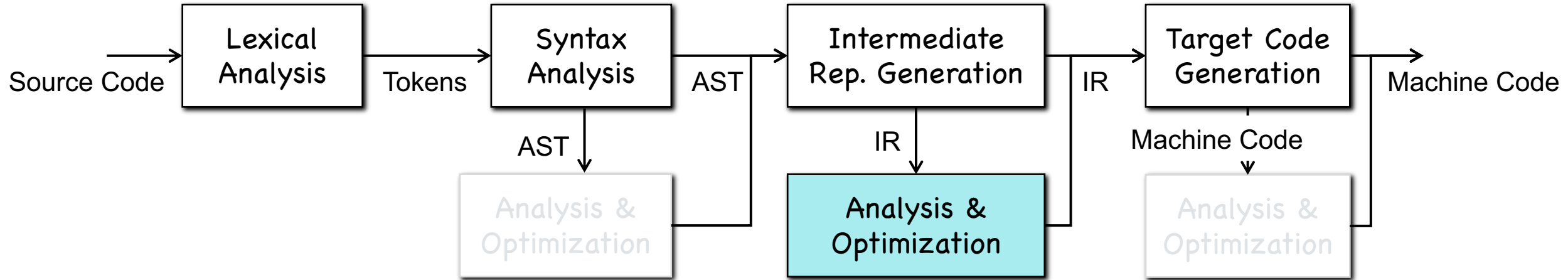
Data Flow Analysis (DFA)

Data Flow Analysis



- IR is friendly to analysis and optimization in compilers
 - Rich source code information like types (compared to machine code)
 - Standard format, control flow graphs, etc. (compared to source code)
 - ...

Data Flow Analysis



- Data flow analysis is a form of static program analysis
 - **Machine-independent compiler optimization**
 - Detection of software errors and vulnerabilities
 - ...

Sources of Optimization

- Redundancy in source code

```
std::vector<int> Vec;
...
for (int K = 0; K < Vec.size();
    ...
}
```

```
struct A { int x; int y; } ;
...
A *Obj = new A;
printf("%d\n", Obj->y);
printf("%d\n", Obj->y);
```

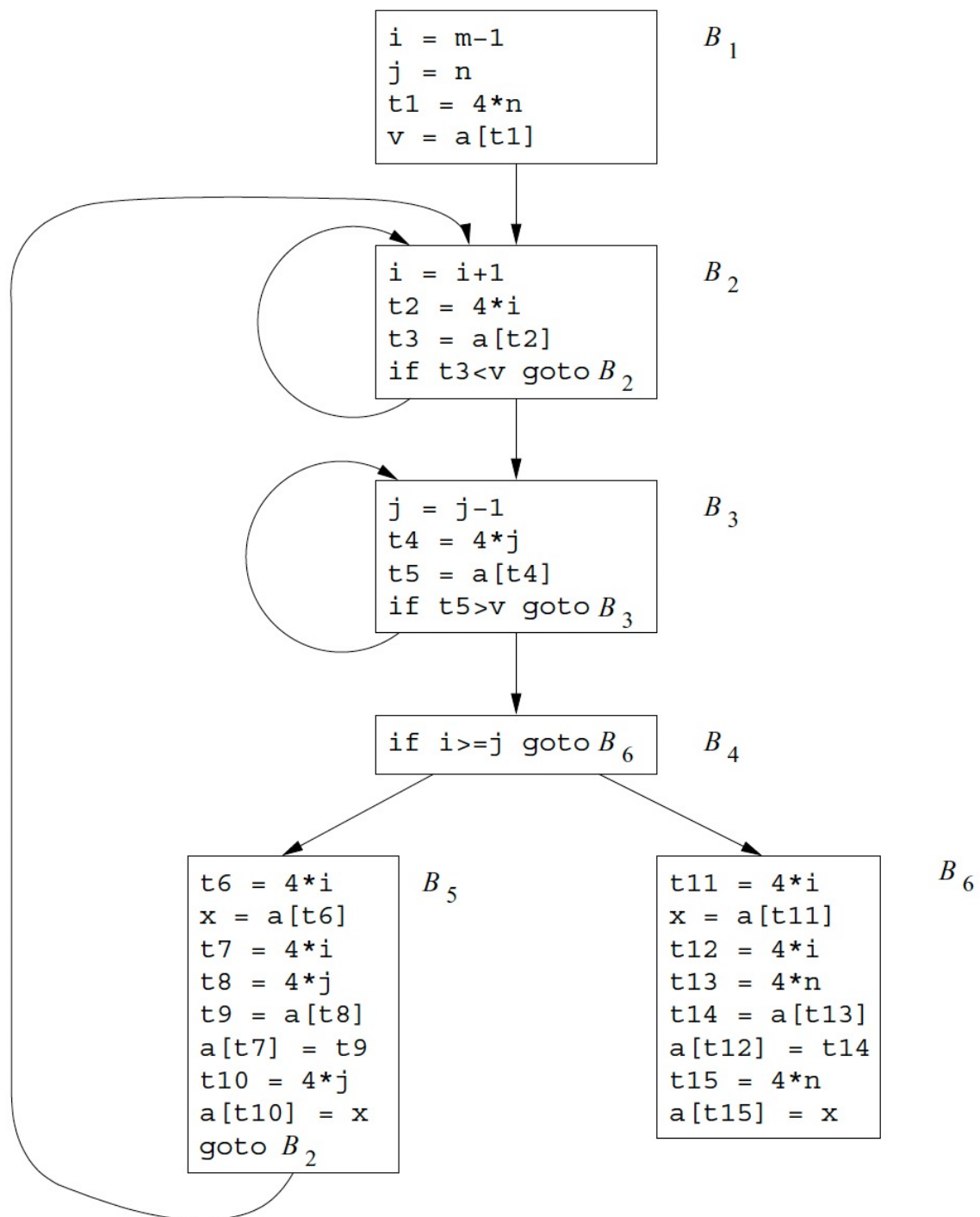
```
; Function Attrs: noline norecurse optnone ssp uwtable mustprogress
define dso_local i32 @main() #0 {
    %1 = alloca i32, align 4
    %2 = alloca %struct.A*, align 8
    store i32 0, i32* %1, align 4
    %3 = call noalias nonnull i8* @_Znwm(i64 8) #3
    %4 = bitcast i8* %3 to %struct.A*
    store %struct.A* %4, %struct.A** %2, align 8
    %5 = load %struct.A*, %struct.A** %2, align 8
    %6 = getelementptr inbounds %struct.A, %struct.A* %5, i32 0, i32 1
    store i32 1, i32* %6, align 4
    %7 = load %struct.A*, %struct.A** %2, align 8
    %8 = getelementptr inbounds %struct.A, %struct.A* %7, i32 0, i32 1
    %9 = load i32, i32* %8, align 4
    %10 = call i32 (i8*, ...) @printf(i8* getelementptr inbounds ([4 x i8], [4 x i8]* @.str, i64 0,
    %11 = load %struct.A*, %struct.A** %2, align 8
    %12 = getelementptr inbounds %struct.A, %struct.A* %11, i32 0, i32 1
    %13 = load i32, i32* %12, align 4
    %14 = call i32 (i8*, ...) @printf(i8* getelementptr inbounds ([4 x i8], [4 x i8]* @.str, i64 0,
    ret i32 0
}
```

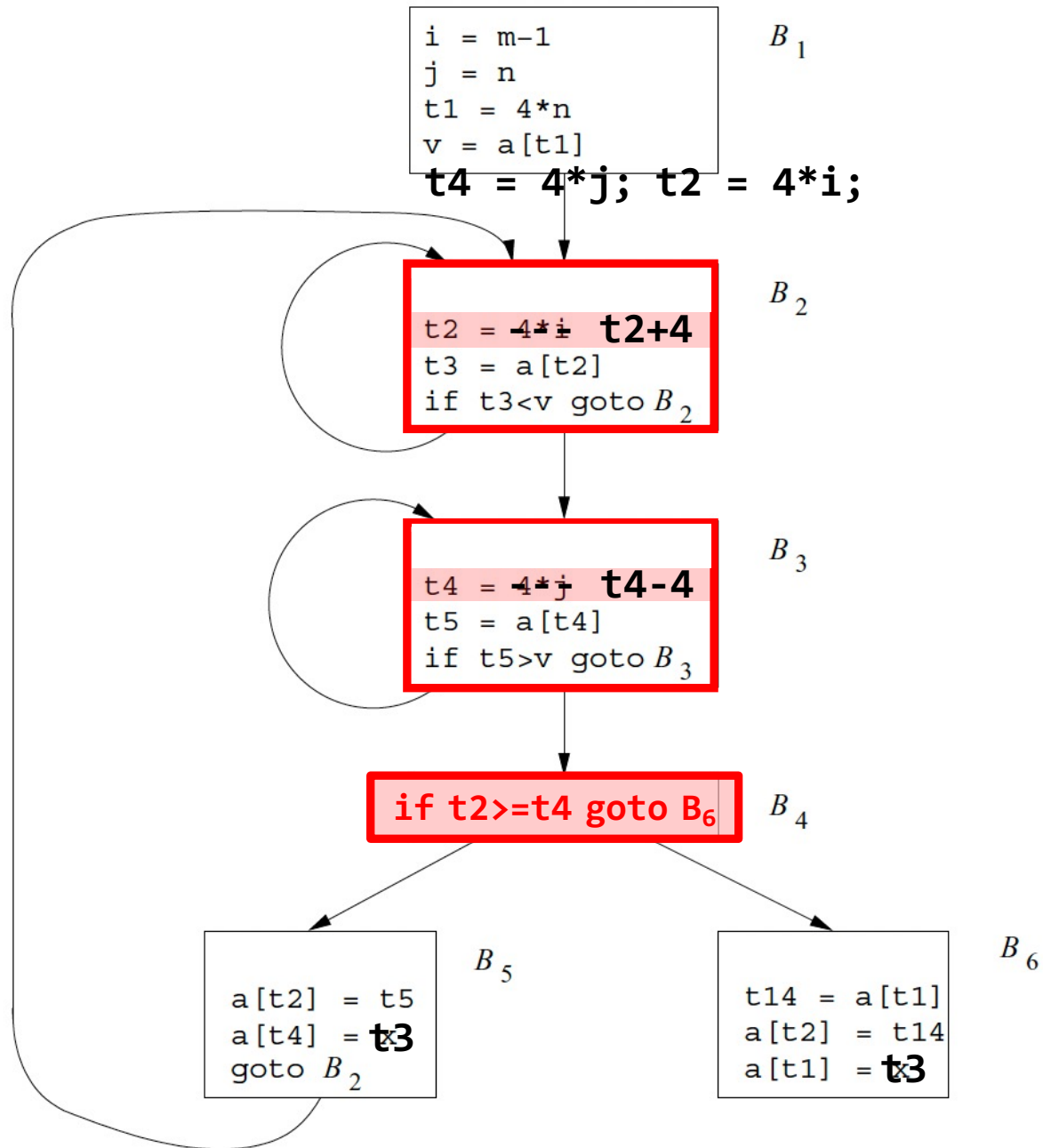
Quick Sort: A Running Example

```
void quicksort(int m, int n)
    /* recursively sorts a[m] through a[n] */
{
    int i, j;
    int v, x;
    if (n <= m) return;
    /* fragment begins here */
    i = m-1; j = n; v = a[n];
    while (1) {
        do i = i+1; while (a[i] < v);
        do j = j-1; while (a[j] > v);
        if (i >= j) break;
        x = a[i]; a[i] = a[j]; a[j] = x; /* swap a[i], a[j] */
    }
    x = a[i]; a[i] = a[n]; a[n] = x; /* swap a[i], a[n] */
    /* fragment ends here */
    quicksort(m, j); quicksort(i+1, n);
}
```

Quick Sort: A Running Example

(1)	i = m-1	(16)	t7 = 4*i
(2)	j = n	(17)	t8 = 4*j
(3)	t1 = 4*n	(18)	t9 = a[t8]
(4)	v = a[t1]	(19)	a[t7] = t9
(5)	i = i+1	(20)	t10 = 4*j
(6)	t2 = 4*i	(21)	a[t10] = x
(7)	t3 = a[t2]	(22)	goto (5)
(8)	if t3<v goto (5)	(23)	t11 = 4*i
(9)	j = j-1	(24)	x = a[t11]
(10)	t4 = 4*j	(25)	t12 = 4*i
(11)	t5 = a[t4]	(26)	t13 = 4*n
(12)	if t5>v goto (9)	(27)	t14 = a[t13]
(13)	if i>=j goto (23)	(28)	a[t12] = t14
(14)	t6 = 4*i	(29)	t15 = 4*n
(15)	x = a[t6]	(30)	a[t15] = x





1. Local common subexpression
2. **Global common subexpression**
3. Copy propagation
4. **Strength reduction**
5. **Induction-variable elimination**
6.

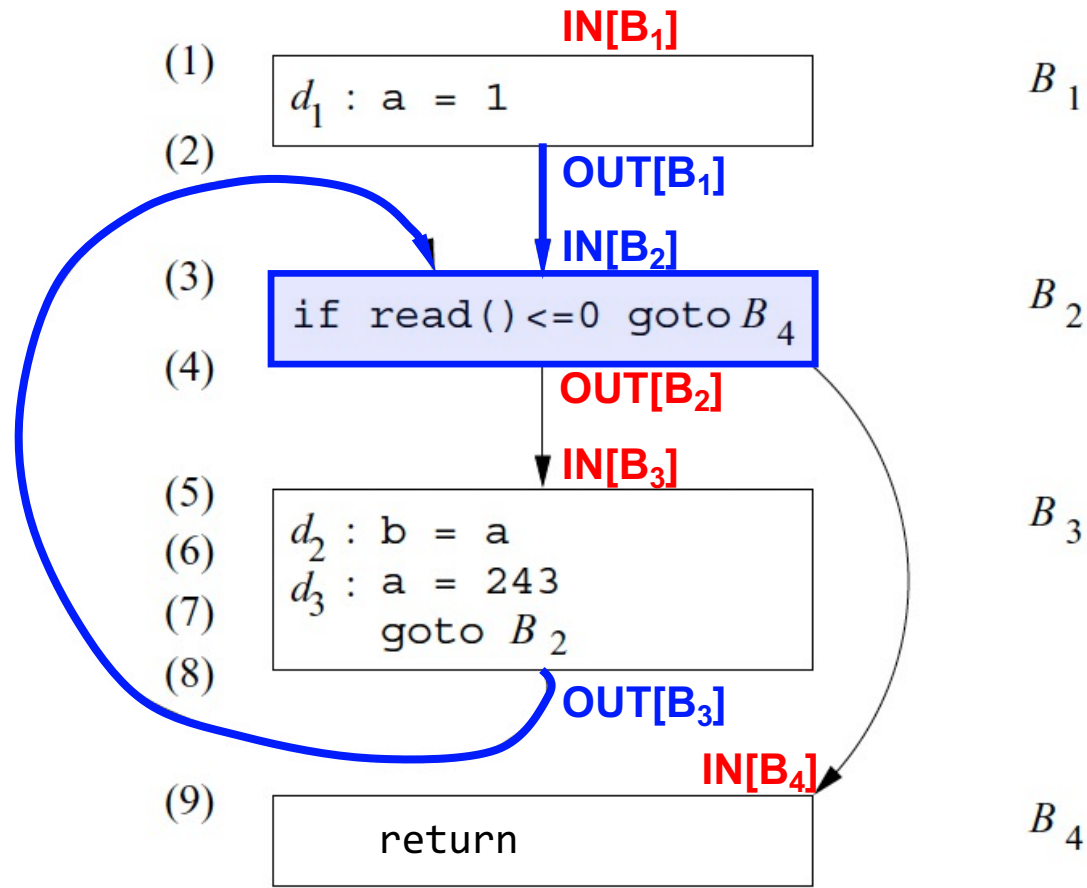
How?

PART I: DFA Scheme

Data Flow Analysis

- Describes how “data flow facts” transfer in a program, which often contains unbounded number of execution paths.
- **Data flow fact** (or value) is an abstraction of all possible concrete program behaviors, *e.g.*,
 - A variable v may have one of the concrete values $\{ 1, 2, 4, 5, 6, \dots \}$
 - Impossible to enumerate all cases in a data flow analysis
 - **Abstraction:** $v = \text{NAC}$;
 - NAC is an abstract value of the concrete values $\{ 1, 2, 4, 5, 6, \dots \}$

Data Flow Analysis



- $IN[B]/OUT[B]$
 - Set of data flow facts before and after a block
- Constraints
 - $OUT[B] = f_B(IN[B])$
 - $IN[B] = \bigwedge_{P \in \text{preds}(B)} OUT[P]$
- $\Rightarrow$ Data Flow Equations

Data Flow Analysis

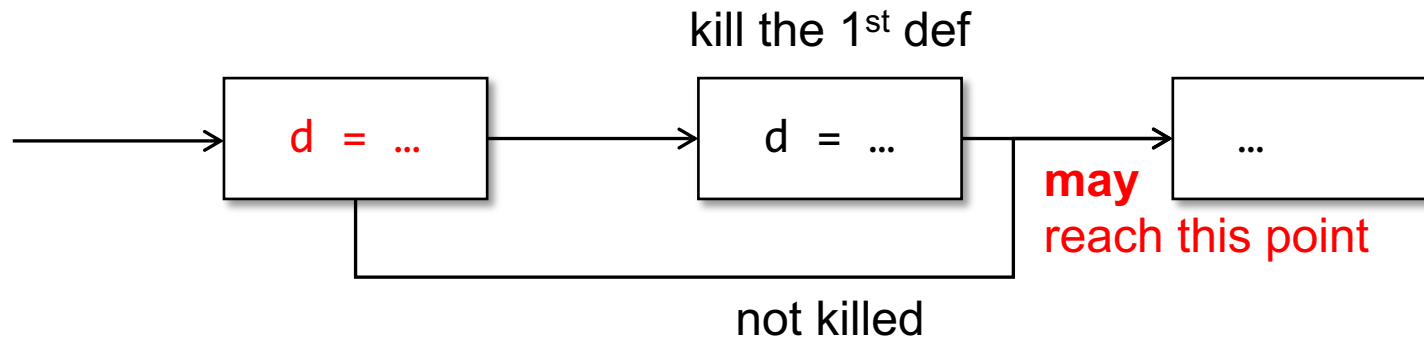
- To perform data flow analysis, define
 - transfer function f for each statement/block
 - merge function $\wedge$ at the joint point of multiple paths
 - the **initial** set of data flow facts for each block
- Data flow analysis produces
 - $IN[B]/OUT[B]$ that include data flow facts at a program point
 - The resulting data flow facts are always true during program execution

The Worklist Algorithm

- **ForEach** basic block B: initialize IN[B] and OUT[B]; **EndFor**
- worklist = set of all basic blocks;
- **While** (!worklist.empty()) **Do**
 - B = worklist.pop();
 - $IN[B] = \bigwedge_{P \in \text{preds}(B)} OUT[P]; \quad OUT[B] = f_B(IN[B])$
 - **If** OUT[B] changed: worklist.push(all B's successors); **EndIf**
- **EndWhile**

Reaching Definitions

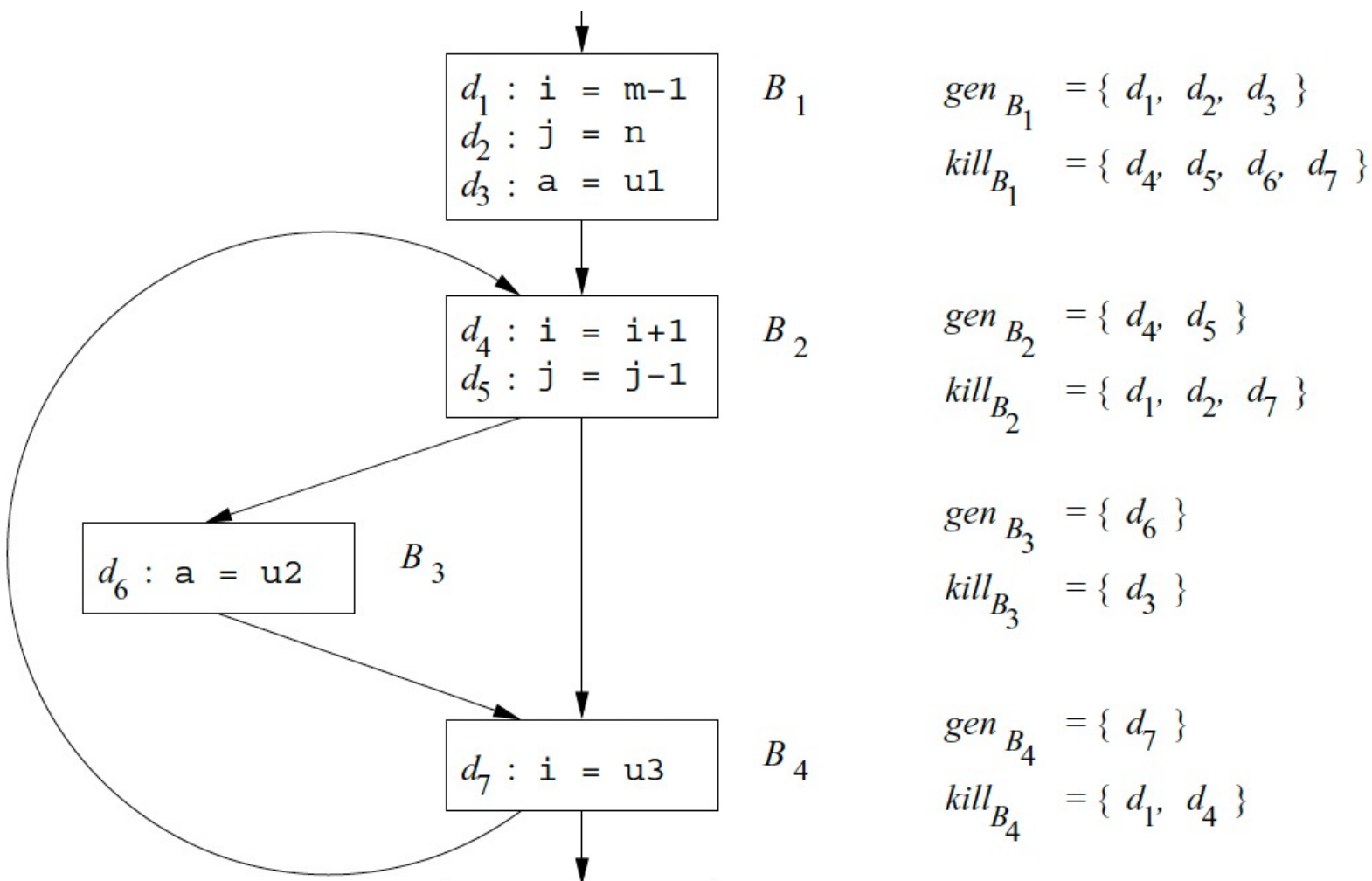
- A definition d **may** reach a program point, if **there is a path from d to the program point** such that **d is not killed along the path**.



Reaching Definitions

- A definition d may reach a program point, if there is a path from d to the program point such that d is not killed along the path.
- $OUT[B] = f_B(IN[B]) = gen_B \cup (IN[B] - kill_B)$
- $IN[B] = \bigwedge_{P \in preds(B)} OUT[P] = \bigcup_{P \in preds(B)} OUT[P]$

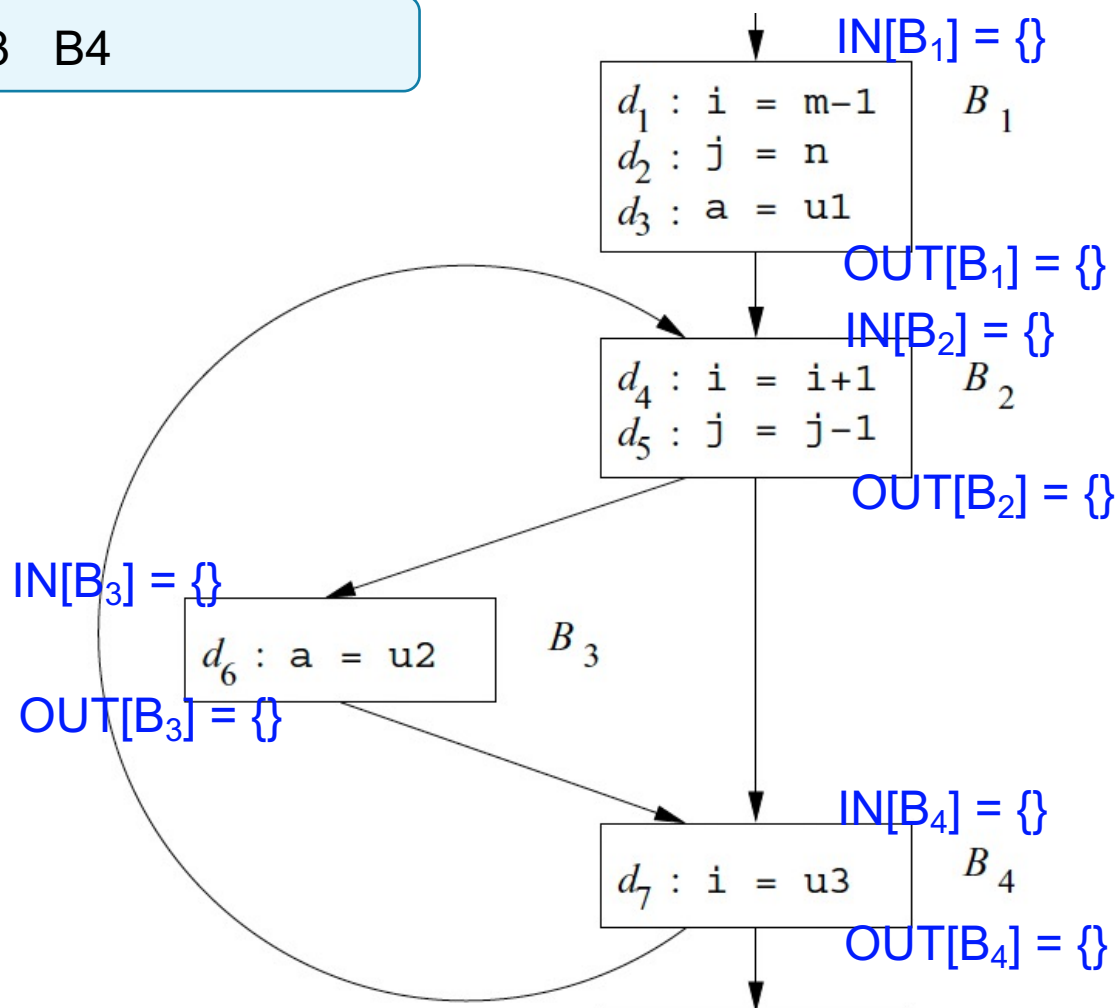
Reaching Definitions



Reaching Definitions

B1 B2 B3 B4

Worklist



$$gen_{B_1} = \{ d_1, d_2, d_3 \}$$

$$kill_{B_1} = \{ d_4, d_5, d_6, d_7 \}$$

$$gen_{B_2} = \{ d_4, d_5 \}$$

$$kill_{B_2} = \{ d_1, d_2, d_7 \}$$

$$gen_{B_3} = \{ d_6 \}$$

$$kill_{B_3} = \{ d_3 \}$$

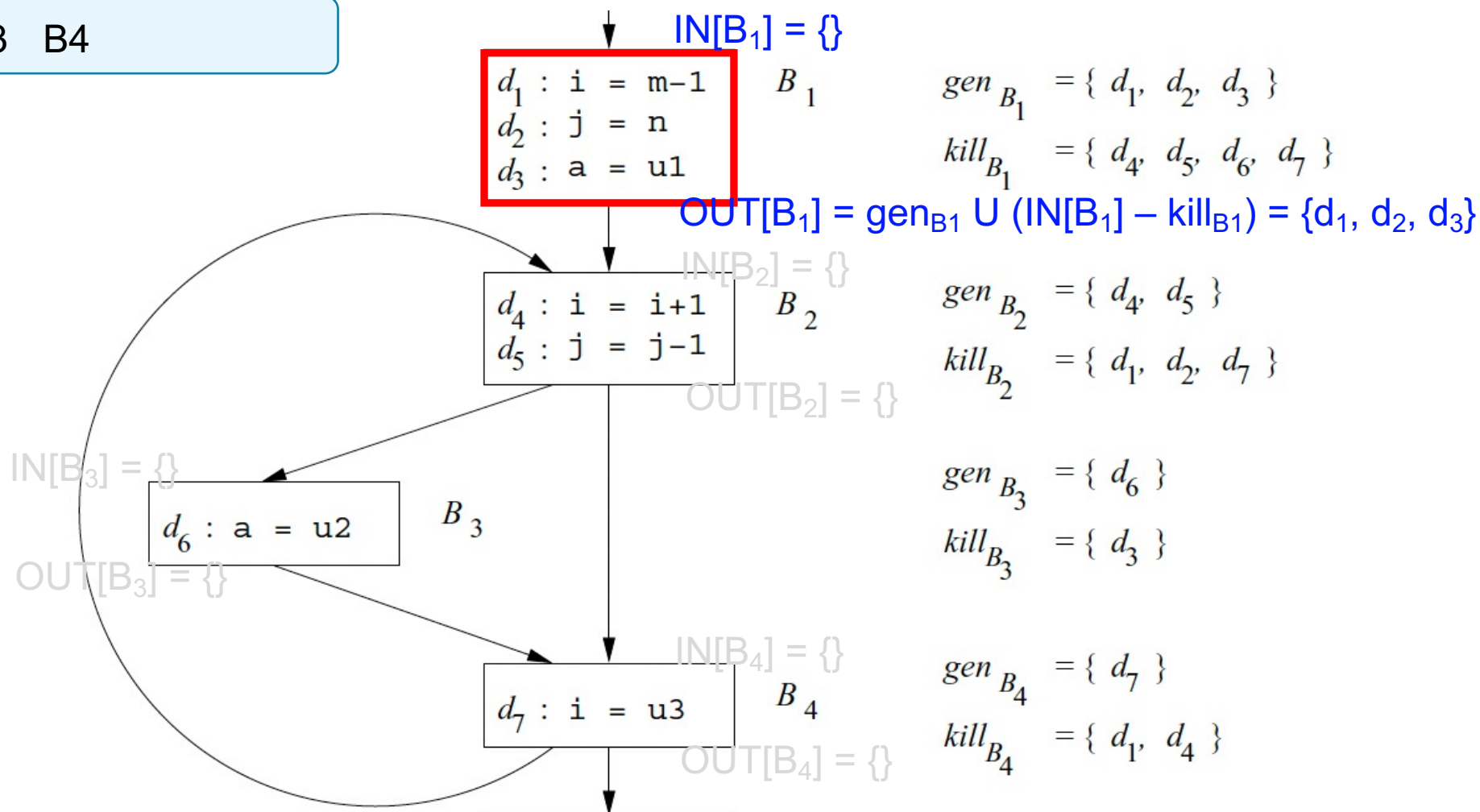
$$gen_{B_4} = \{ d_7 \}$$

$$kill_{B_4} = \{ d_1, d_4 \}$$

Reaching Definitions

B1 B2 B3 B4

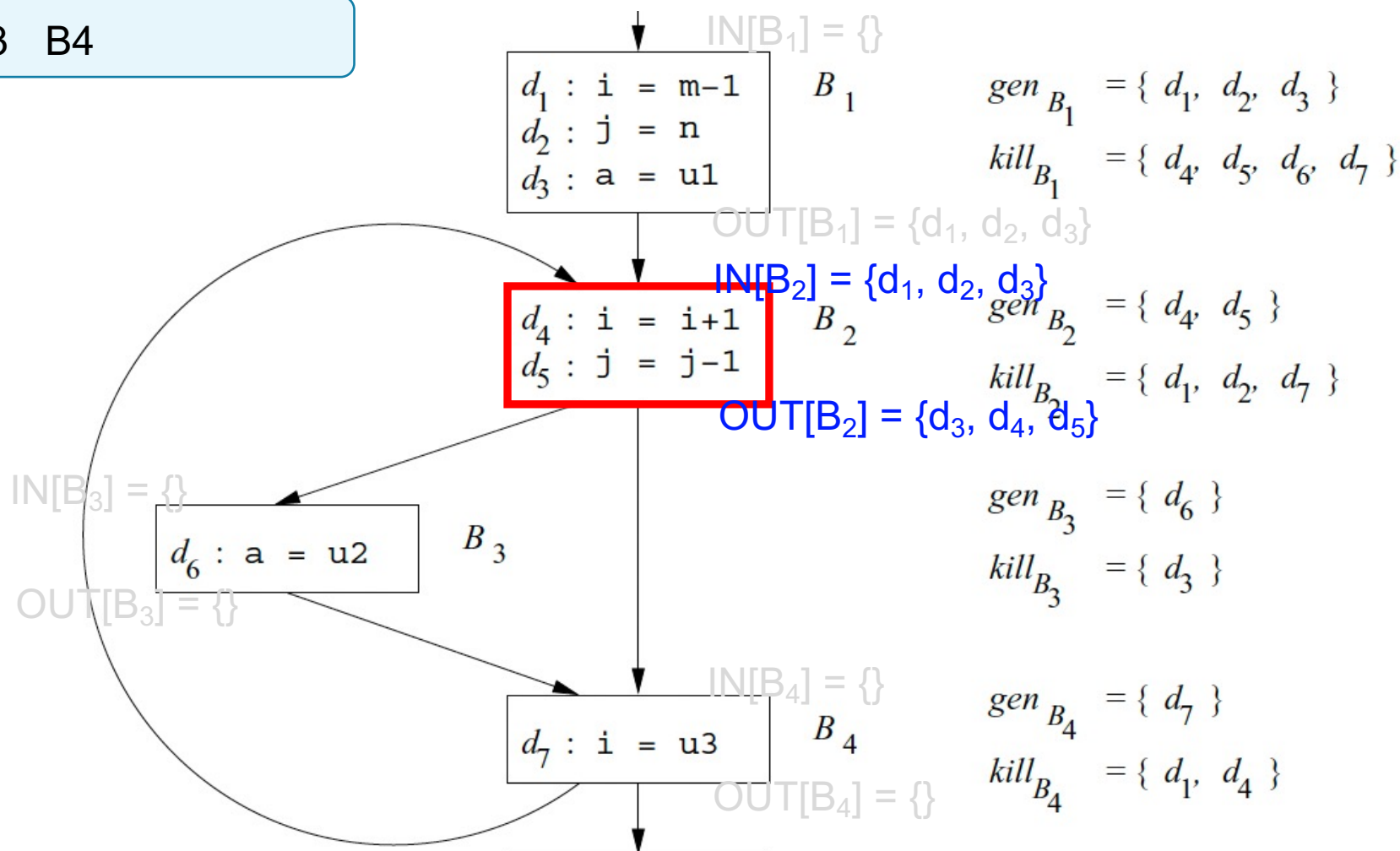
Worklist



Reaching Definitions

B1 B2 B3 B4

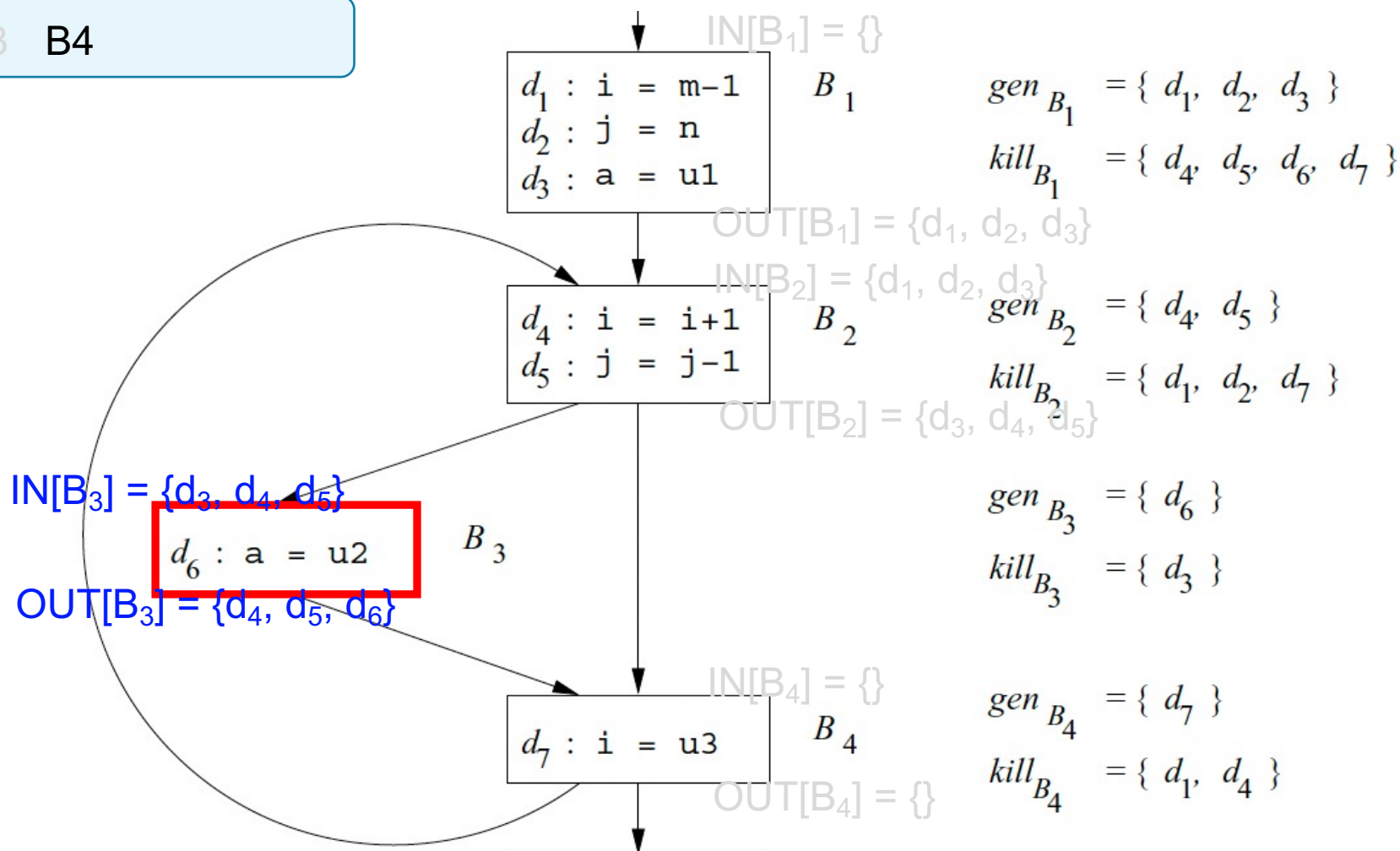
Worklist



Reaching Definitions

B1 B2 B3 B4

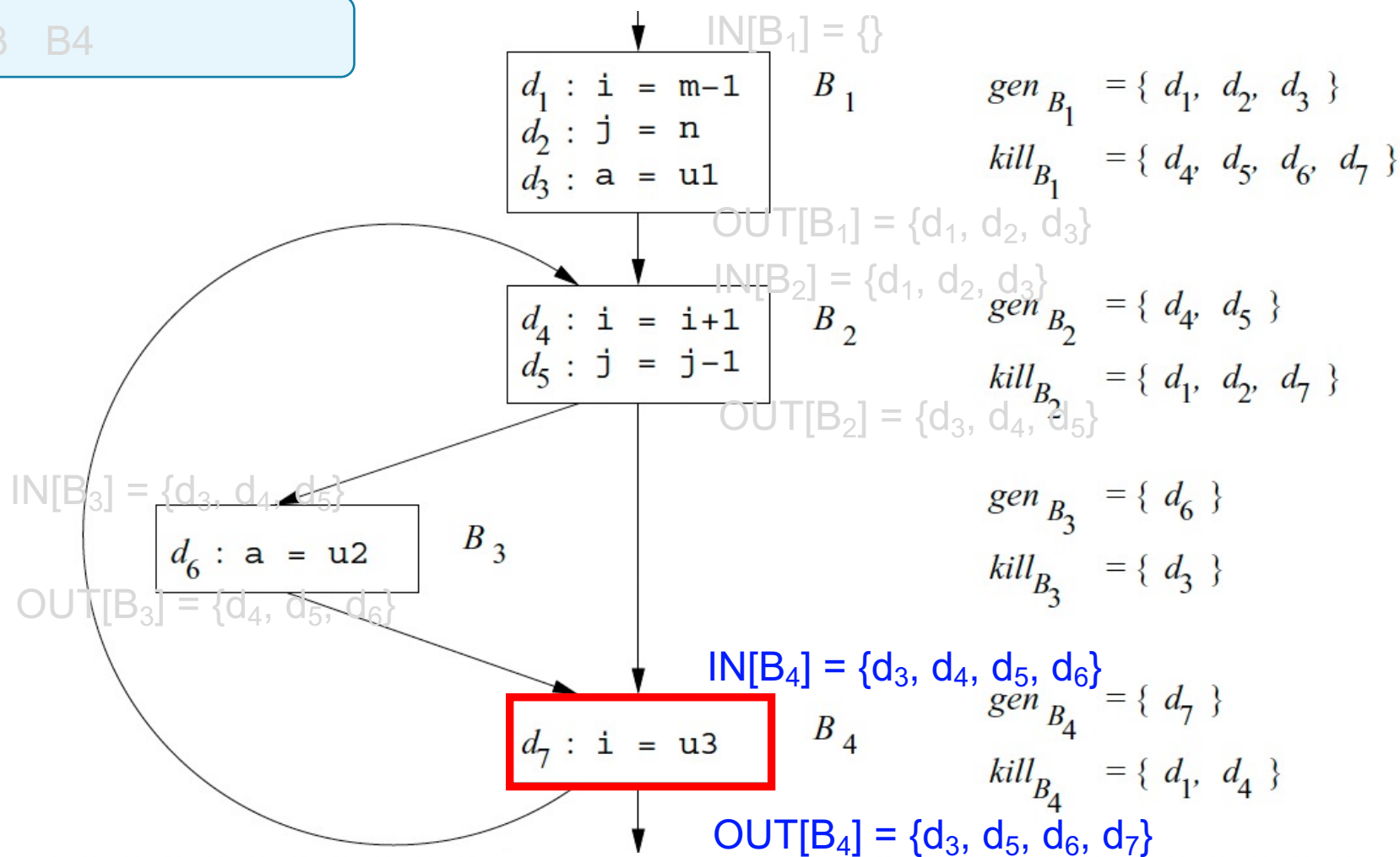
Worklist



Reaching Definitions

B1 B2 B3 B4

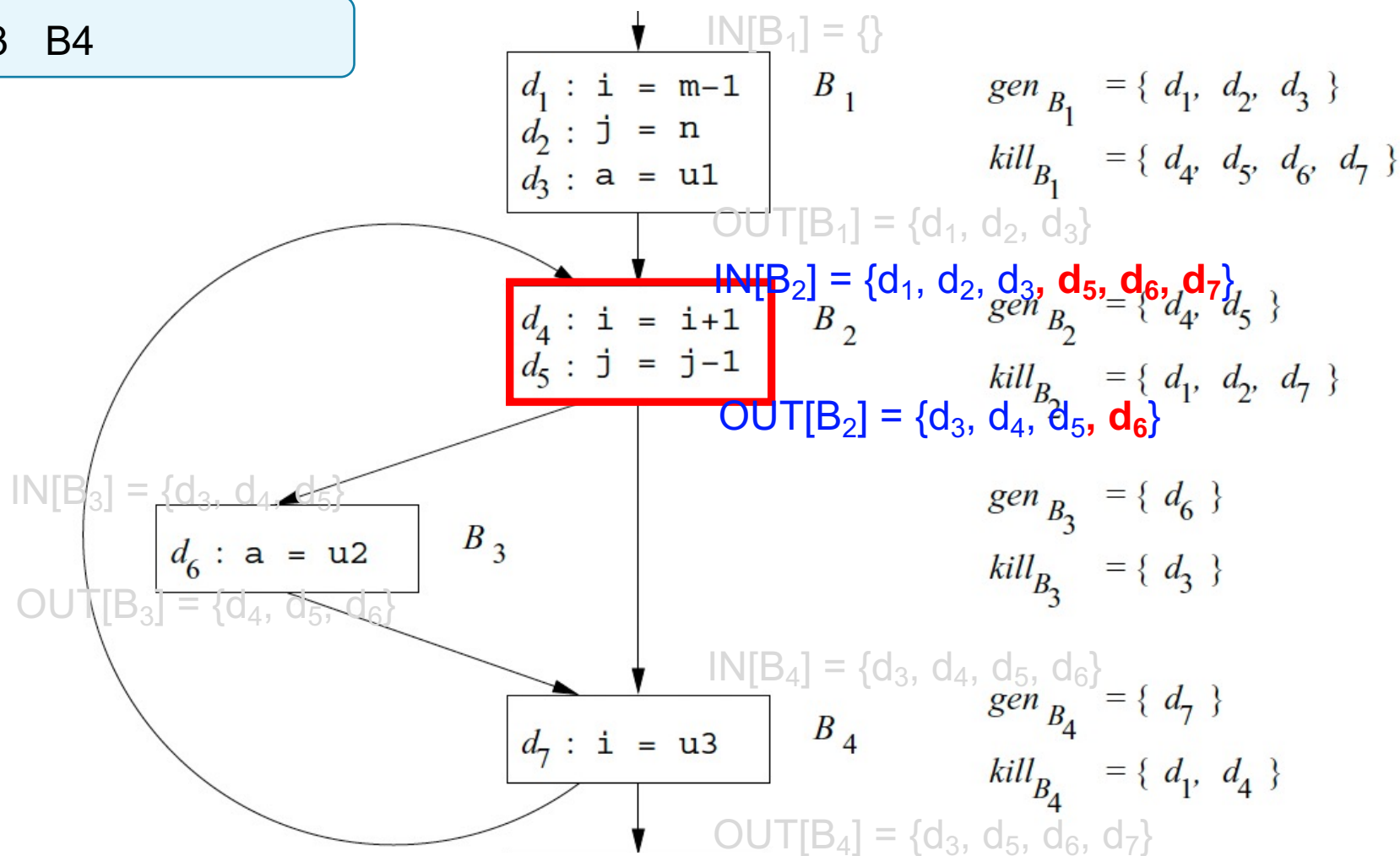
Worklist



Reaching Definitions

B1 B2 B3 B4

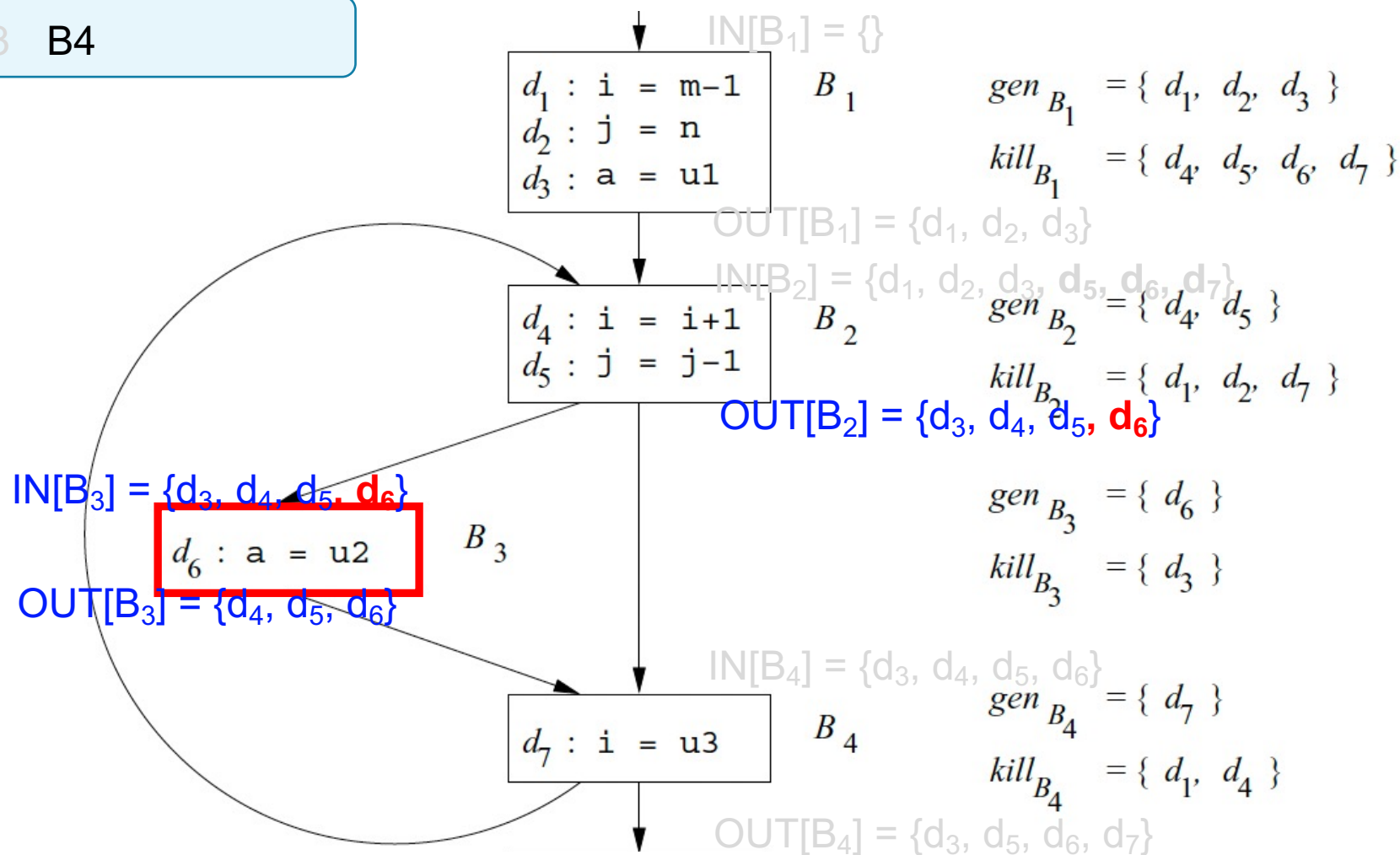
Worklist



Reaching Definitions

B1 B2 B3 B4

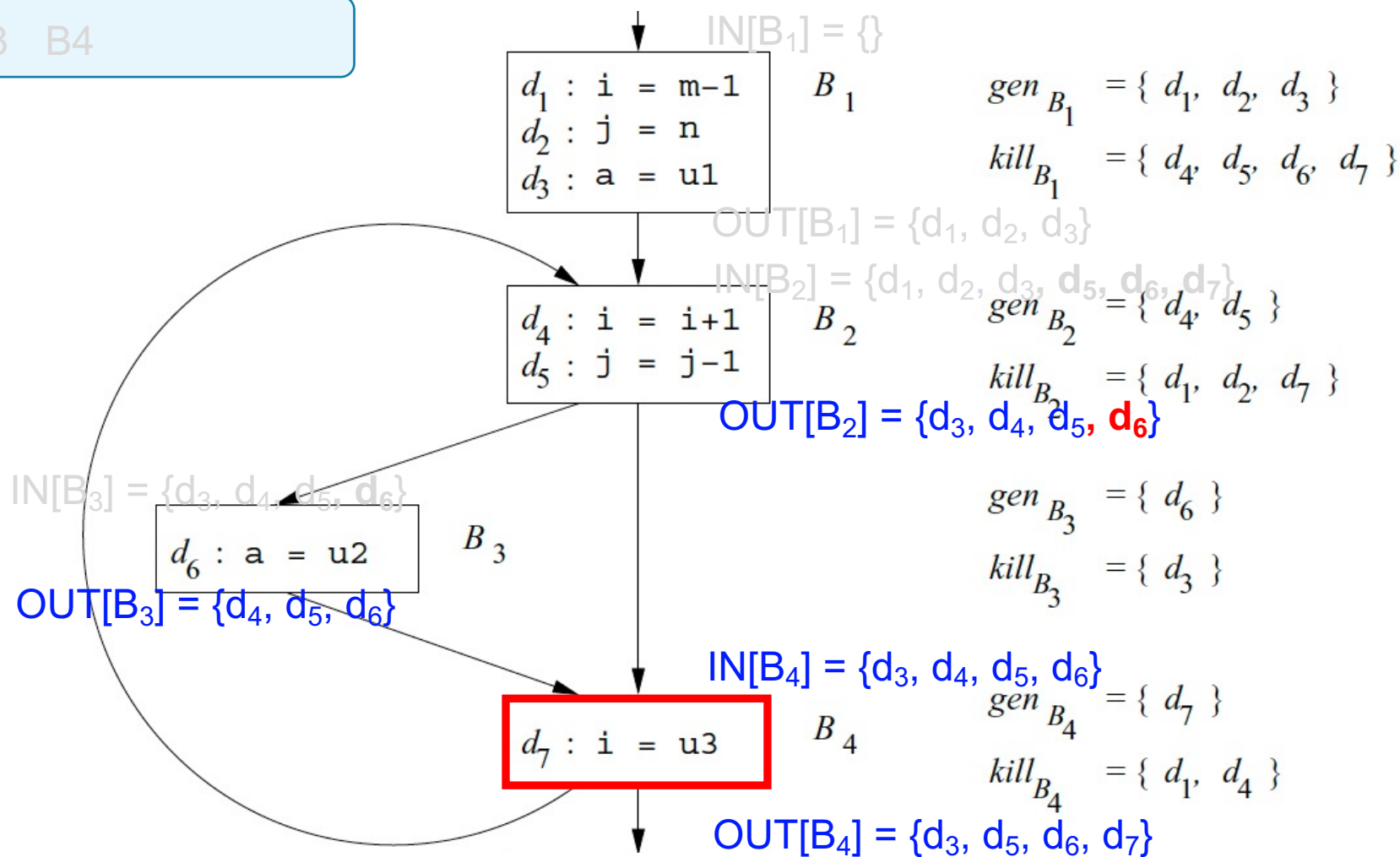
Worklist



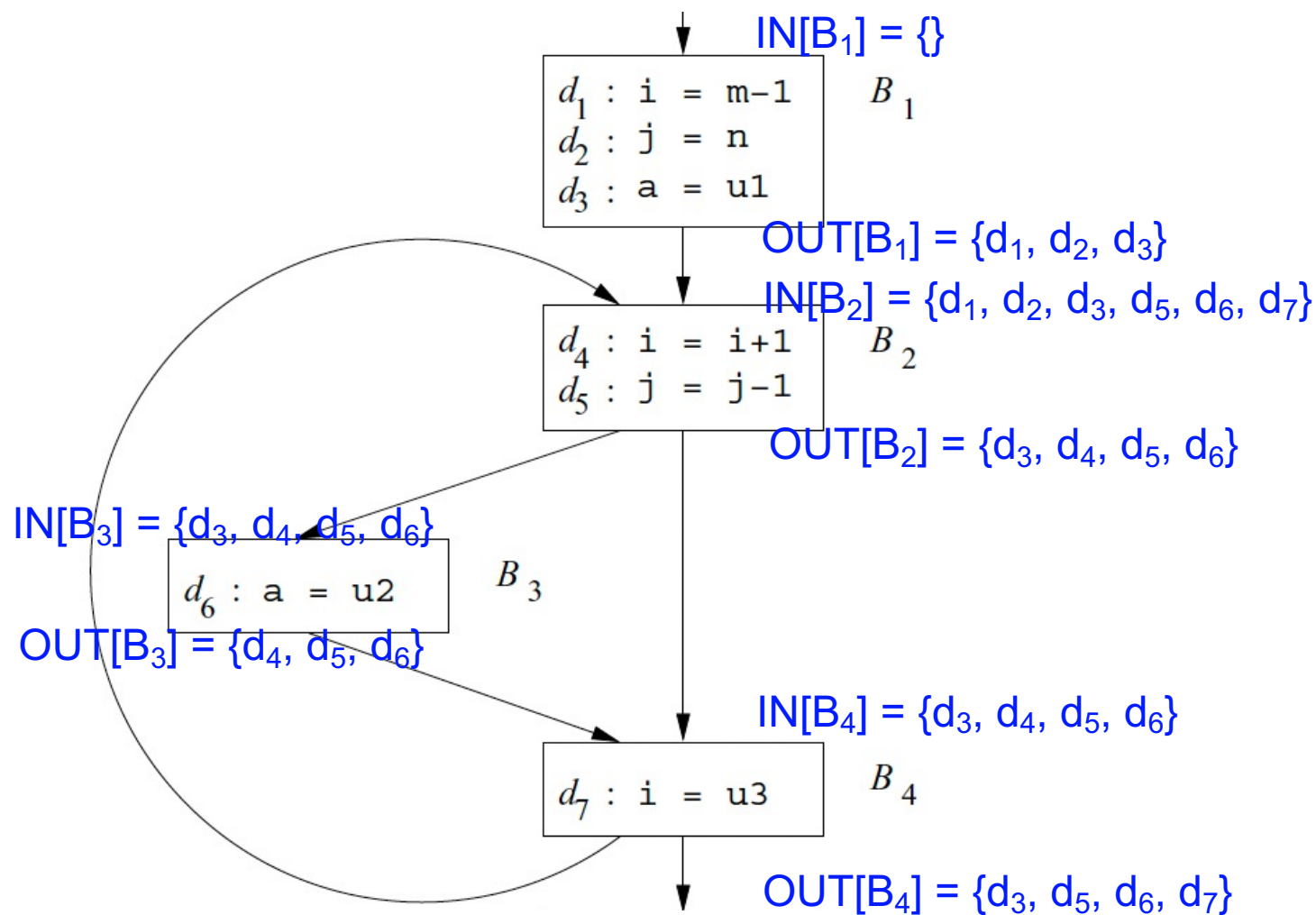
Reaching Definitions

B1 B2 B3 B4

Worklist

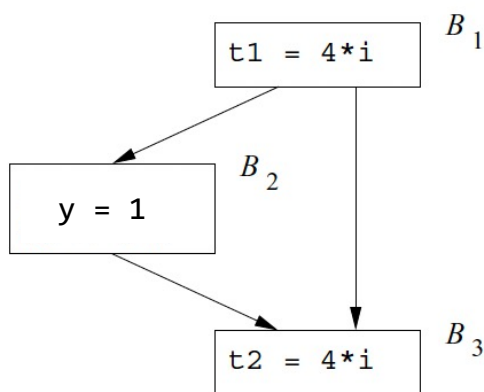


Reaching Definitions

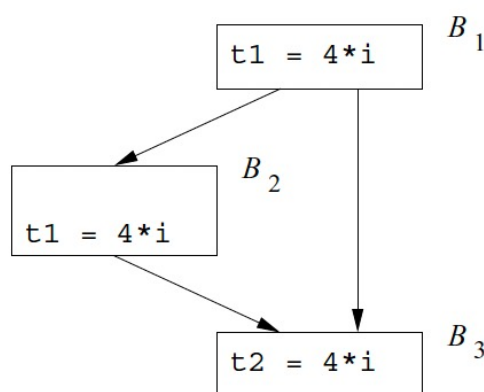


Available Expressions

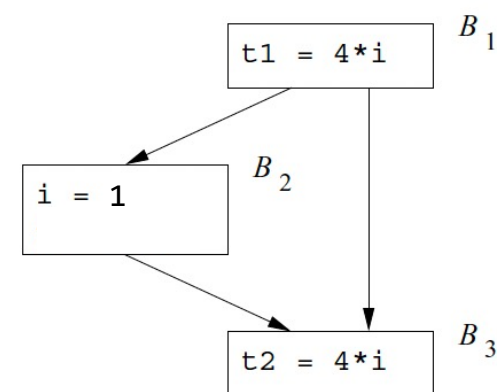
- An expression $x + y$ is available at a point p if
 - every path from the entry node to p evaluates $x + y$, and
 - after the last such evaluation prior to reaching p , there are no subsequent assignments to x or y .



$4 * i$ is available before B_3



$4 * i$ is available before B_3



$4 * i$ is **NOT** available before B_3

Available Expressions

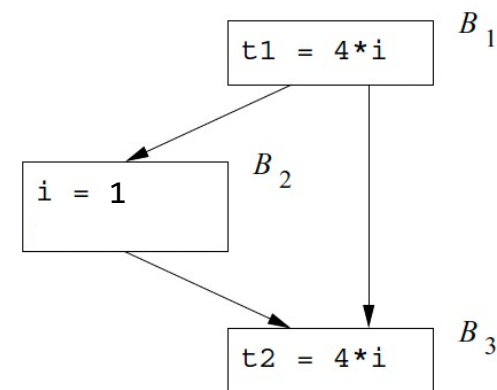
- An expression $x + y$ is available at a point p if
 - every path from the entry node to p evaluates $x + y$, and
 - after the last such evaluation prior to reaching p , there are no subsequent assignments to x or y .

- The equations

$$\text{OUT}[B] = e_gen_B \cup (\text{IN}[B] - e_kill_B)$$

$$\text{IN}[B] = \bigcap_{P \text{ a predecessor of } B} \text{OUT}[P].$$

Why not $\cup$?



$4 * i$ is **NOT** available before B_3

Available Expressions

- An expression $x + y$ is available at a point p if
 - every path from the entry node to p evaluates $x + y$, and
 - after the last such evaluation prior to reaching p , there are no subsequent assignments to x or y .

- The equations

$$\text{OUT}[B] = e_gen_B \cup (\text{IN}[B] - e_kill_B)$$

$$\text{IN}[B] = \bigcap_{P \text{ a predecessor of } B} \text{OUT}[P].$$

Statement	Available Expressions
	$\emptyset$
$a = b + c$	$\{b + c\}$
$b = a - d$	$\{a - d\}$
$c = b + c$	$\{a - d\}$
$d = a - d$	$\emptyset$

Live Variable Analysis

- We wish to know if the value of a variable x at point p can be used along the path starting from p .

The values of y , z at this point are **live** as we can use them in the next statement.

The value of x at this point is **dead** because x will be redefined and the old value cannot be used any longer.

$x = y + z$

Live Variable Analysis

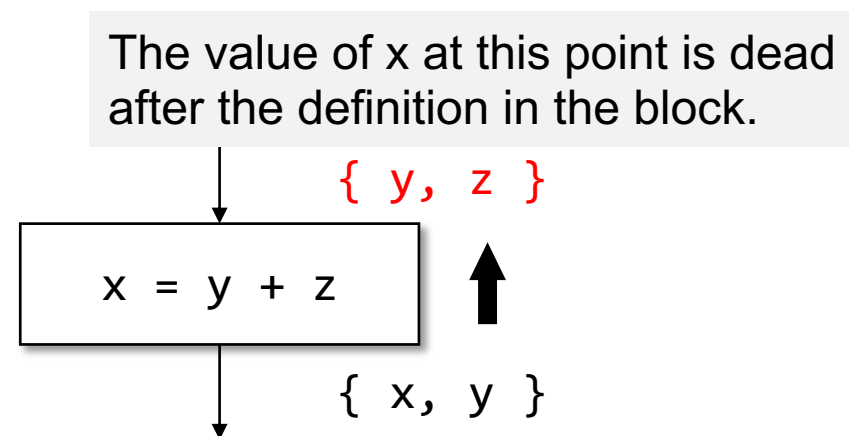
- **Application:** Register Allocation.
- A dead value
 - After a variable is revised, its previous value (in the register) is dead
 - We need to reload its value from the memory to the register
- A live value
 - A variable is not re-defined
 - After loading to a register, the value in the register is valid
 - No need to reload the value from the memory

Live Variable Analysis

- We wish to know if the value of a variable x at point p can be used along the path starting from p .
- **Backward data flow analysis.**
 - The same as forward analysis except for the direction.

$$IN[B] = use_B \cup (OUT[B] - def_B)$$

$$OUT[B] = \bigcup_{S \text{ a successor of } B} IN[S]$$

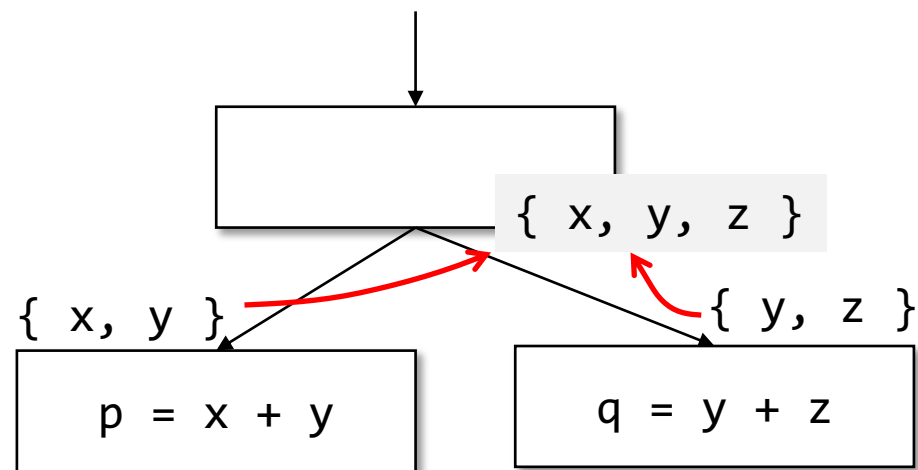


Live Variable Analysis

- We wish to know if the value of a variable x at point p can be used along the path starting from p .
- **Backward data flow analysis.**
 - The same as forward analysis except for the direction.

$$IN[B] = use_B \cup (OUT[B] - def_B)$$

$$OUT[B] = \bigcup_{S \text{ a successor of } B} IN[S]$$



Summary

	Reaching Definitions	Live Variables	Available Expressions
Domain	Sets of definitions	Sets of variables	Sets of expressions
Direction	Forwards	Backwards	Forwards
Transfer function	$gen_B \cup (x - kill_B)$	$use_B \cup (x - def_B)$	$e_gen_B \cup (x - e_kill_B)$
Meet ($\wedge$)	$\cup$	$\cup$	$\cap$
Initialize	$OUT[B] = \emptyset$	$IN[B] = \emptyset$	$OUT[B] = U$

Question-1

- **ForEach** basic block B: initialize IN[B] and OUT[B]; **EndFor**
- worklist = set of all basic blocks;

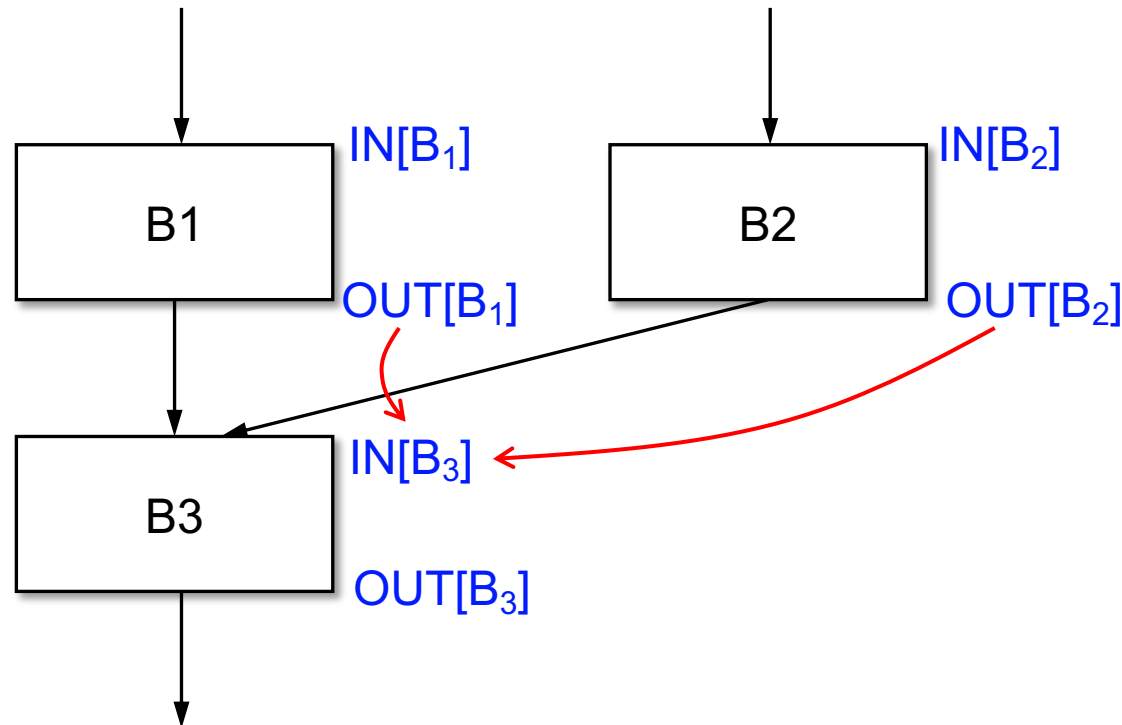
- **While** (!worklist.empty()) **Do**

What's the best order of blocks?

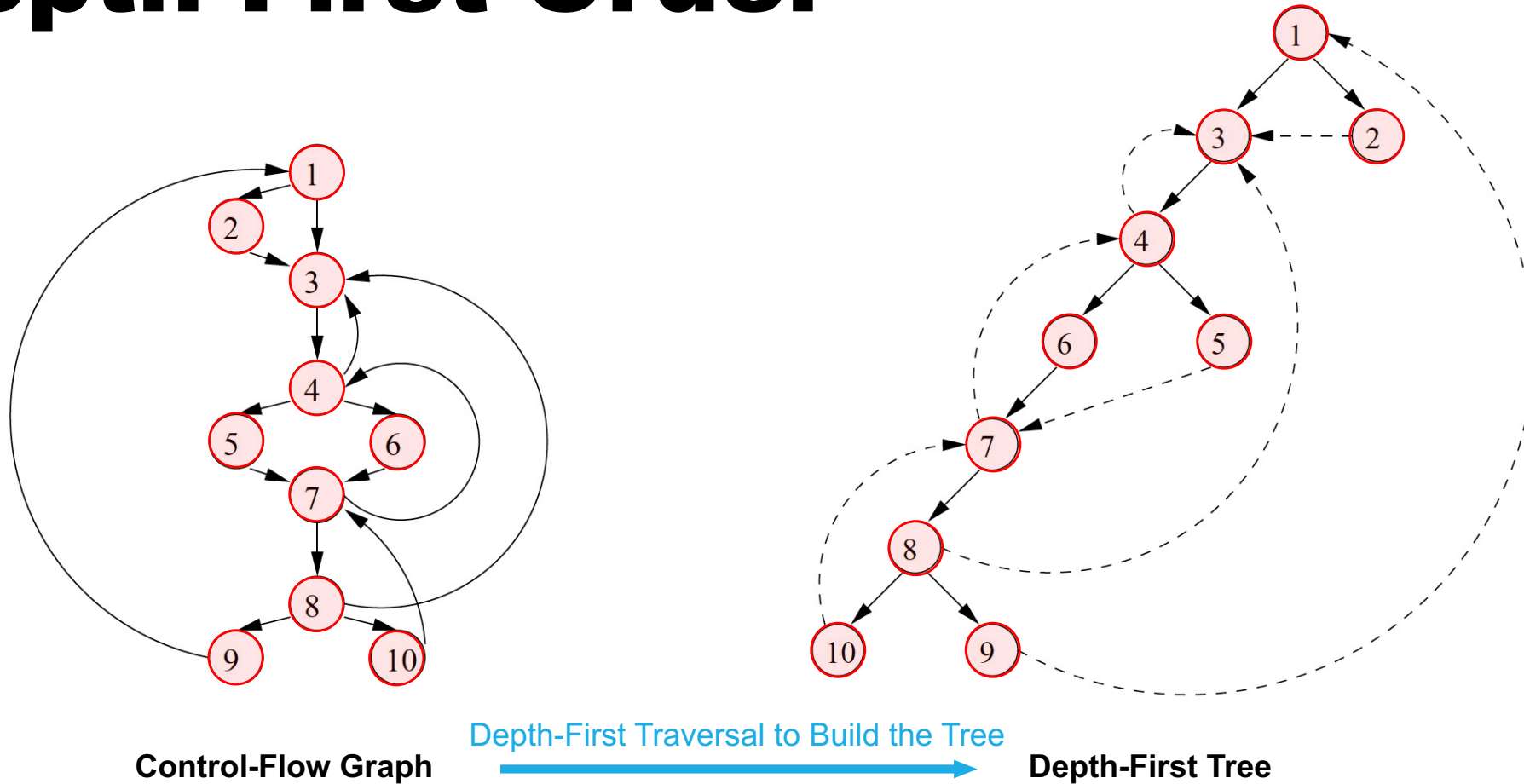
- **B = worklist.pop();**
- $IN[B] = \bigwedge_{P \in \text{preds}(B)} OUT[P]; \quad OUT[B] = f_B(IN[B])$
- **If** OUT[B] changed: worklist.push(all B's successors); **EndIf**
- **EndWhile**

What's the Best Order of Blocks

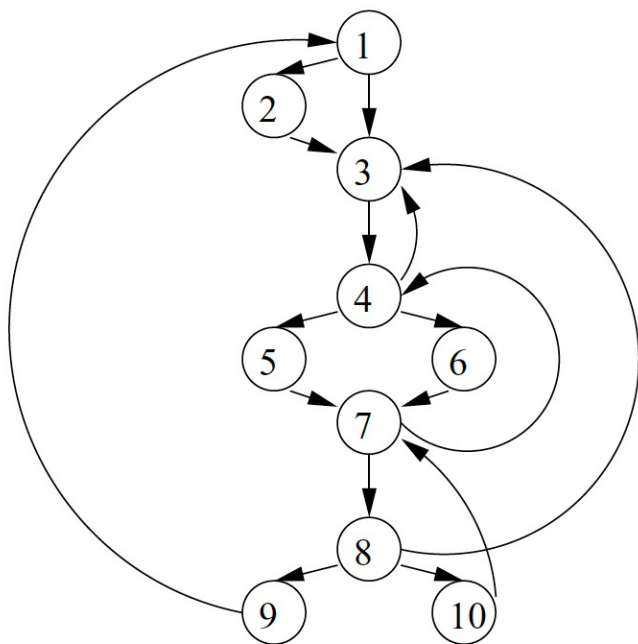
- If we analyze B3 before B1 and B2, B3 may be analyzed again after B1 and B2 are analyzed



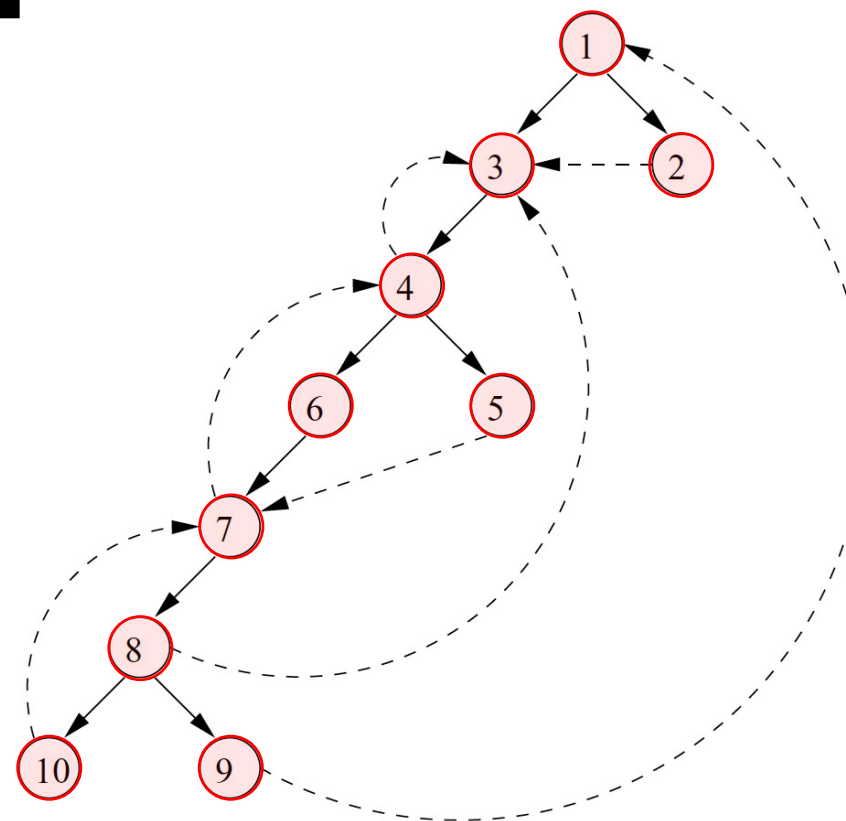
Depth-First Order



Depth-First Order



Control-Flow Graph



Depth-First Tree

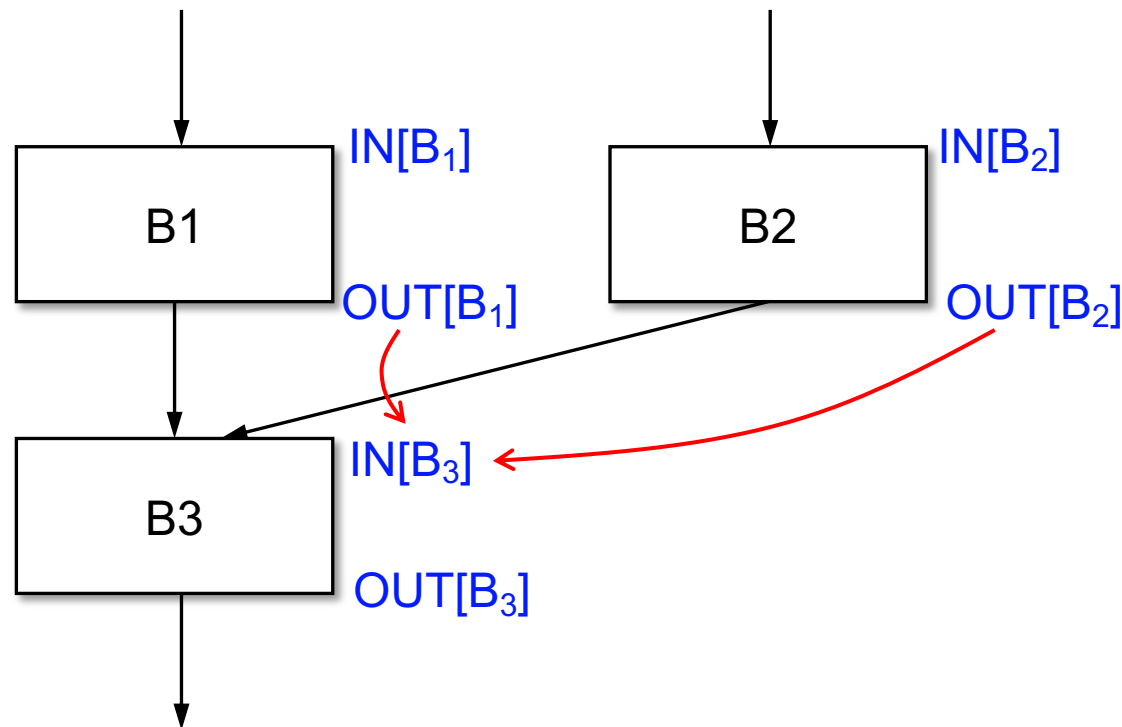
Depth-First Order is the Reversed Post-Order Sequence of the Depth-First Tree

Post-Order of the Tree: 10, 9, 8, 7, 6, 5, 4, 3, 2, 1

Depth-First Order: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Depth-First Order

- It is exactly the topological order when the graph is acyclic
- It provides a topo-like order for cyclic graphs



Question-II

- **ForEach** basic block B: initialize IN[B] and OUT[B]; **EndFor**
- worklist = set of all basic blocks;

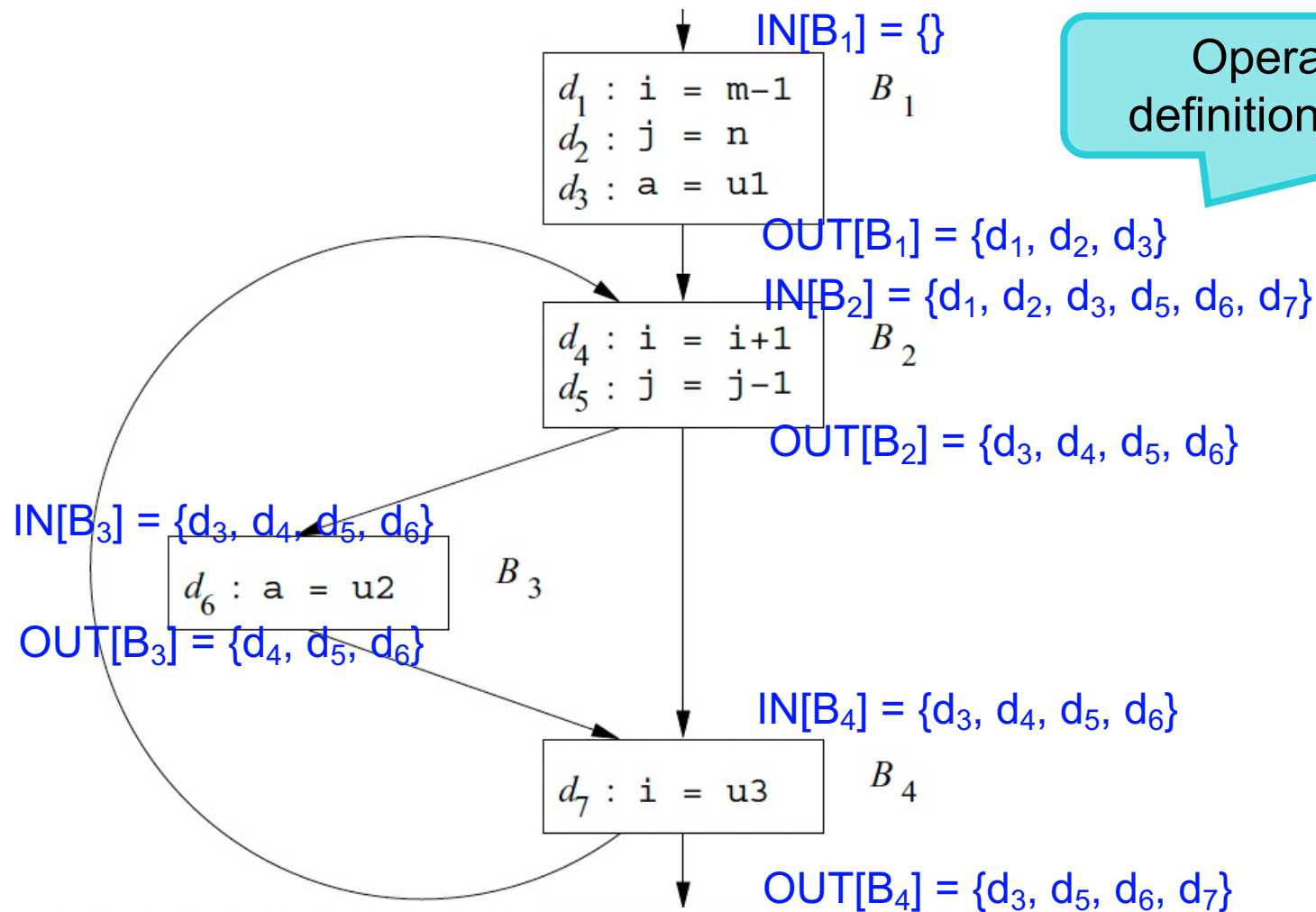
Can the loop terminate?

- **While** (!worklist.empty()) **Do**

Terminable if for each block B:
IN[B] and OUT[B] is **monotonically**
increased or decreased to a limit.

- B = worklist.pop();
- $IN[B] = \bigwedge_{P \in \text{preds}(B)} OUT[P]; \quad OUT[B] = f_B(IN[B])$
- **If** OUT[B] changed: worklist.push(all B's successors); **EndIf**
- **EndWhile**

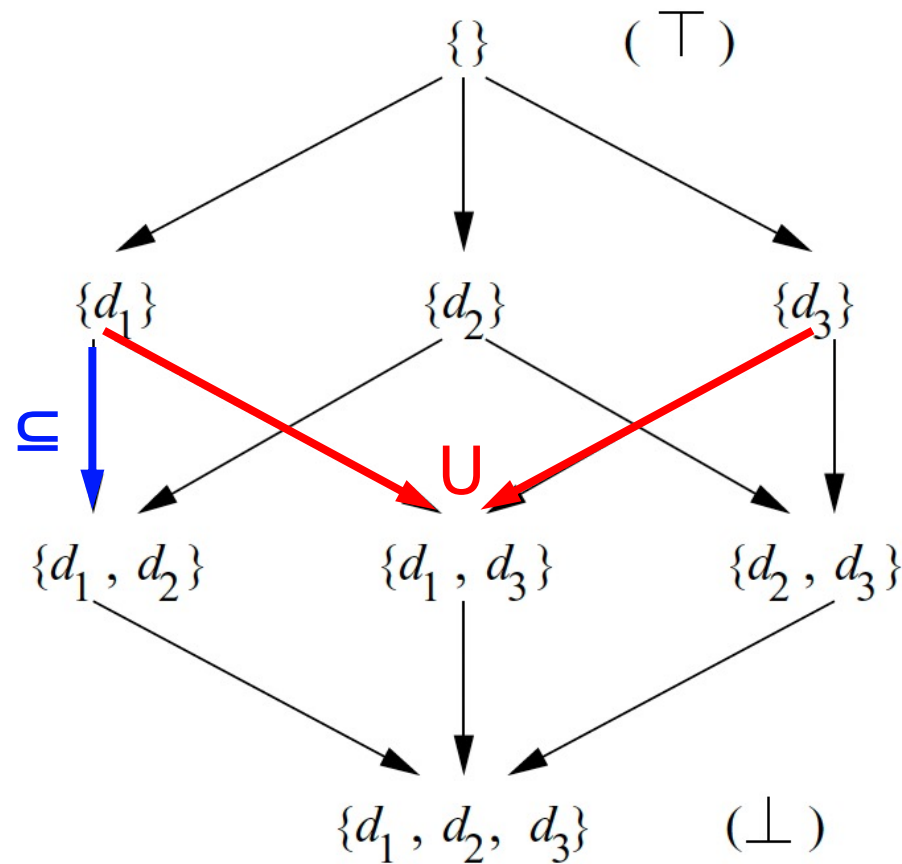
Recall: Reaching Definitions



Operations over subsets of definitions, forming a semi-lattice

Semi-Lattice

- A semilattice is a set V with a meet operation $\wedge$, such that
- (1) $x \wedge x = x$
- (2) $x \wedge y = y \wedge x$
- (3) $x \wedge (y \wedge z) = (x \wedge y) \wedge z$
- (4) $x \wedge \top = x$
- (5) $\perp \wedge x = \perp$



Semi-Lattice & Monotonicity

- Data Flow Equations

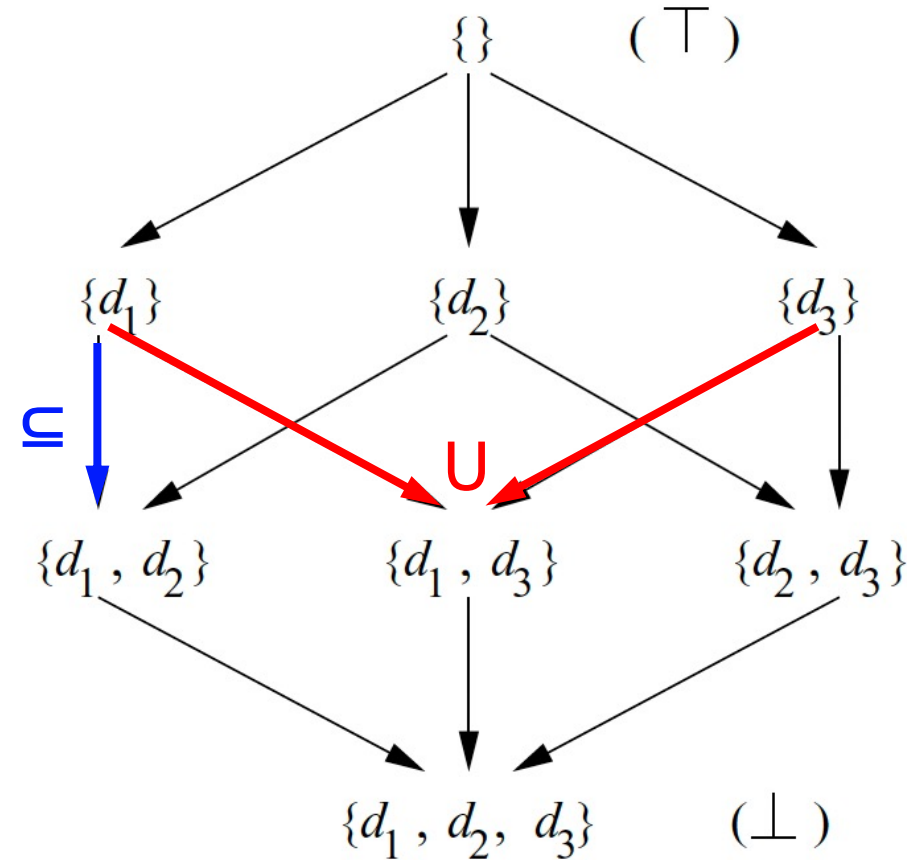
- $IN[B] = \bigcup_{P \in \text{preds}(B)} OUT[P]$ ↗

- $OUT[B] = f_B(IN[B])$ ↗

Monotonicity:

- $x \subseteq y$
- $\Rightarrow \text{gen} \cup (x - \text{kill}) \subseteq \text{gen} \cup (y - \text{kill})$
- $\Rightarrow f_B(x) \subseteq f_B(y)$

The semi-lattice has a finite height.



Question-II

- **ForEach** basic block B: initialize IN[B] and OUT[B]; **EndFor**
- worklist = set of all basic blocks;

- **While** (!worklist.empty()) **Do**

- B = worklist.pop();

- $IN[B] = \bigwedge_{P \in \text{preds}(B)} OUT[P]; \quad OUT[B] = f_B(IN[B])$

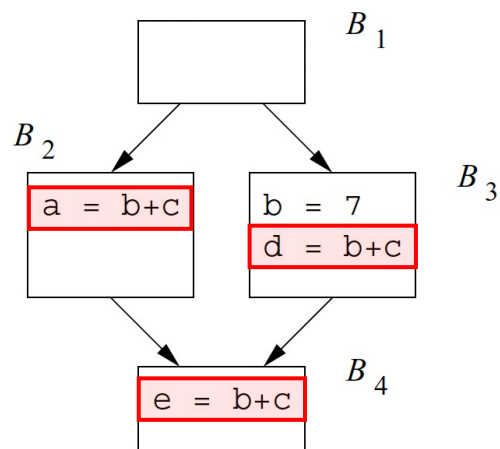
- **If** OUT[B] changed: worklist.push(all B's successors); **EndIf**

- **EndWhile**

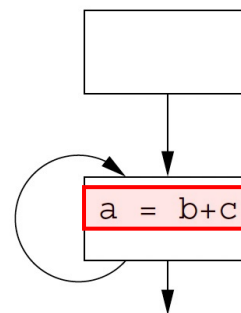
If the semilattice of the framework is monotone and of finite height, then the algorithm is guaranteed to converge.

PART II: Partial Redundancy

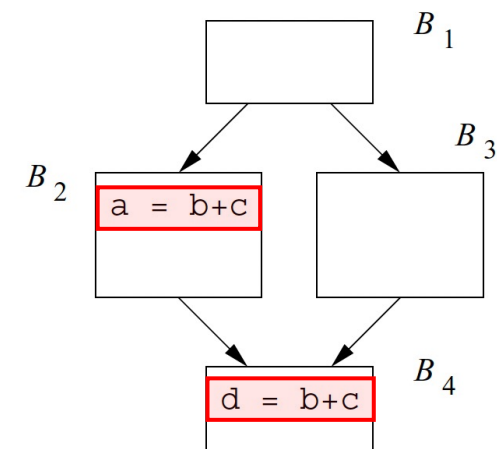
Sources of Redundancy



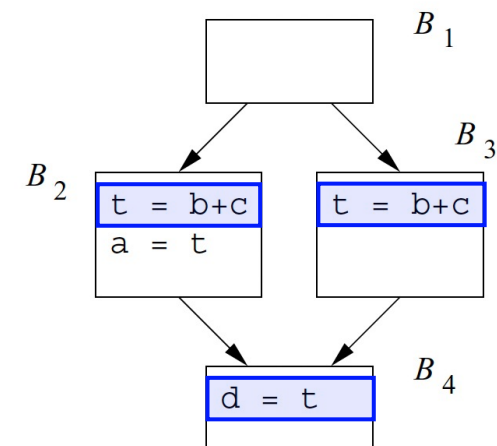
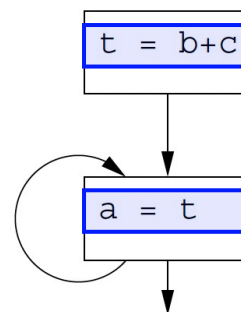
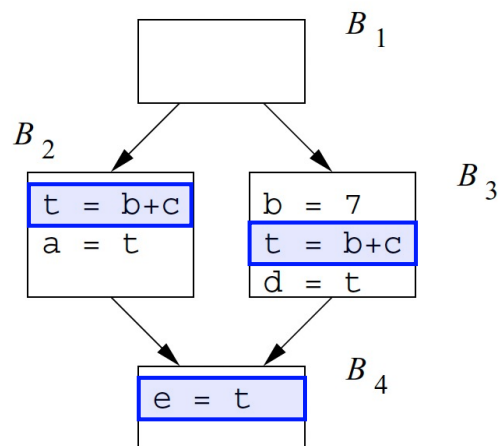
1. Common Expression



2. Loop Invariant Motion

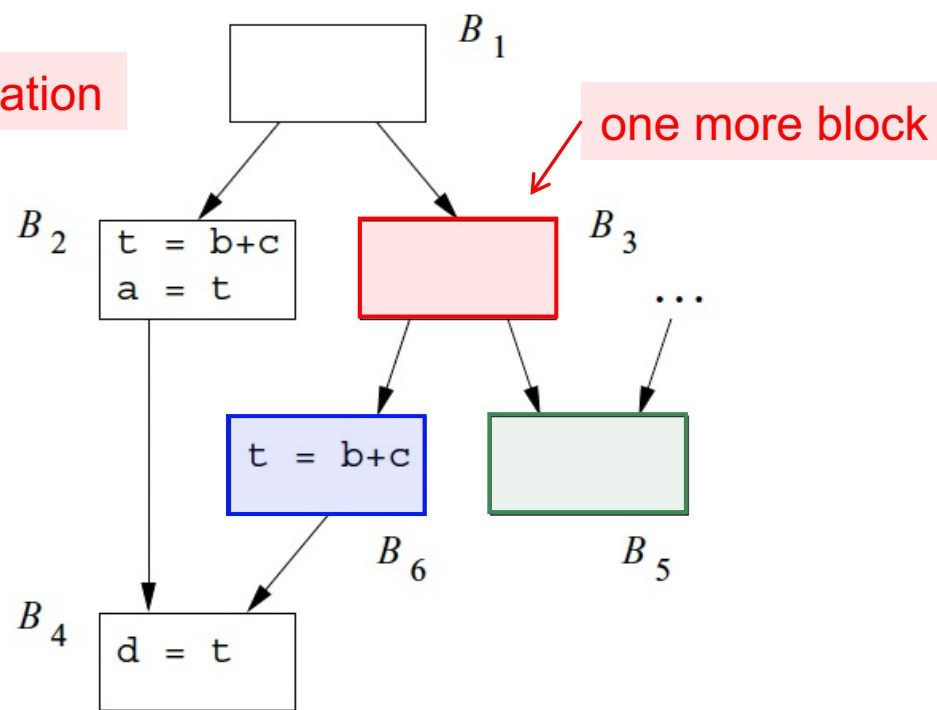
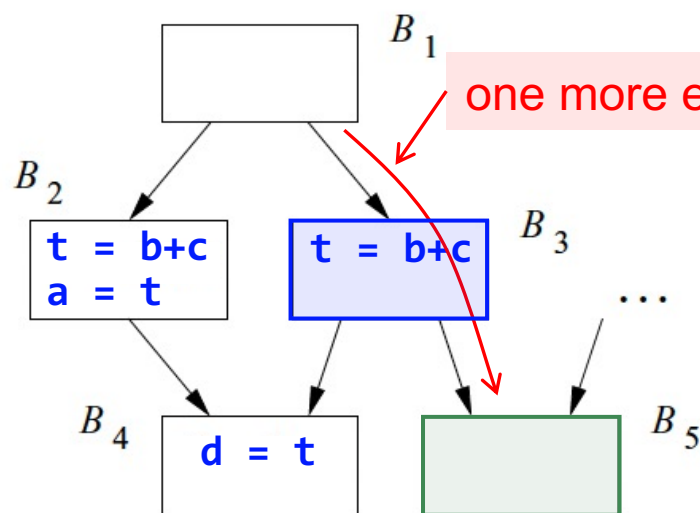


3. Partial Redundancy



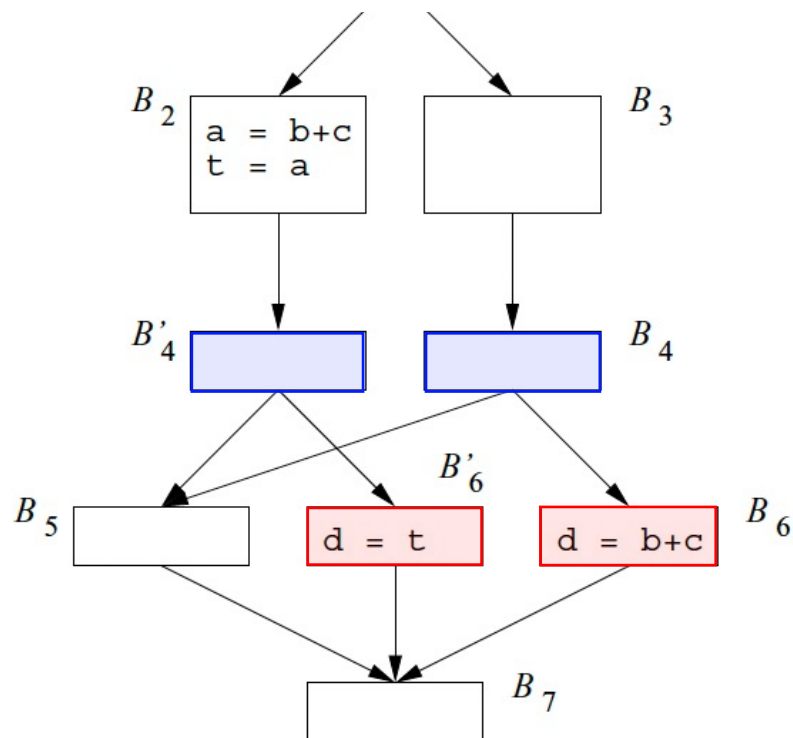
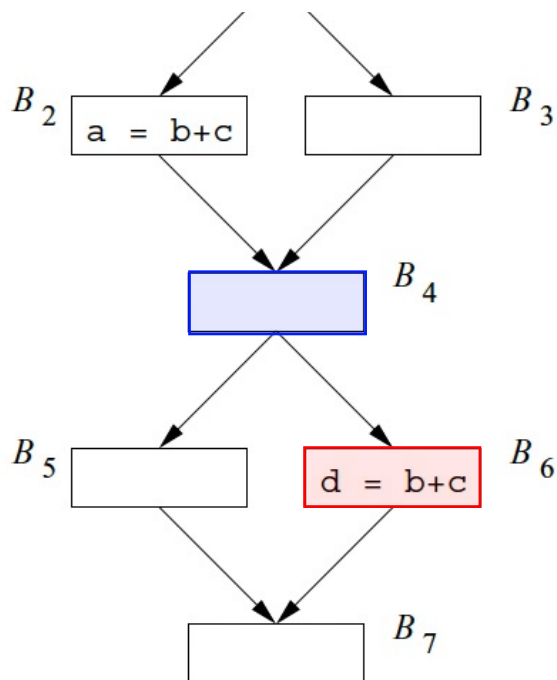
Eliminating All Redundancy?

- Not Possible!
- Unless we are allowed to create new blocks or duplicate code.



Eliminating All Redundancy?

- Not Possible!
- Unless we are allowed to create new blocks or duplicate code.



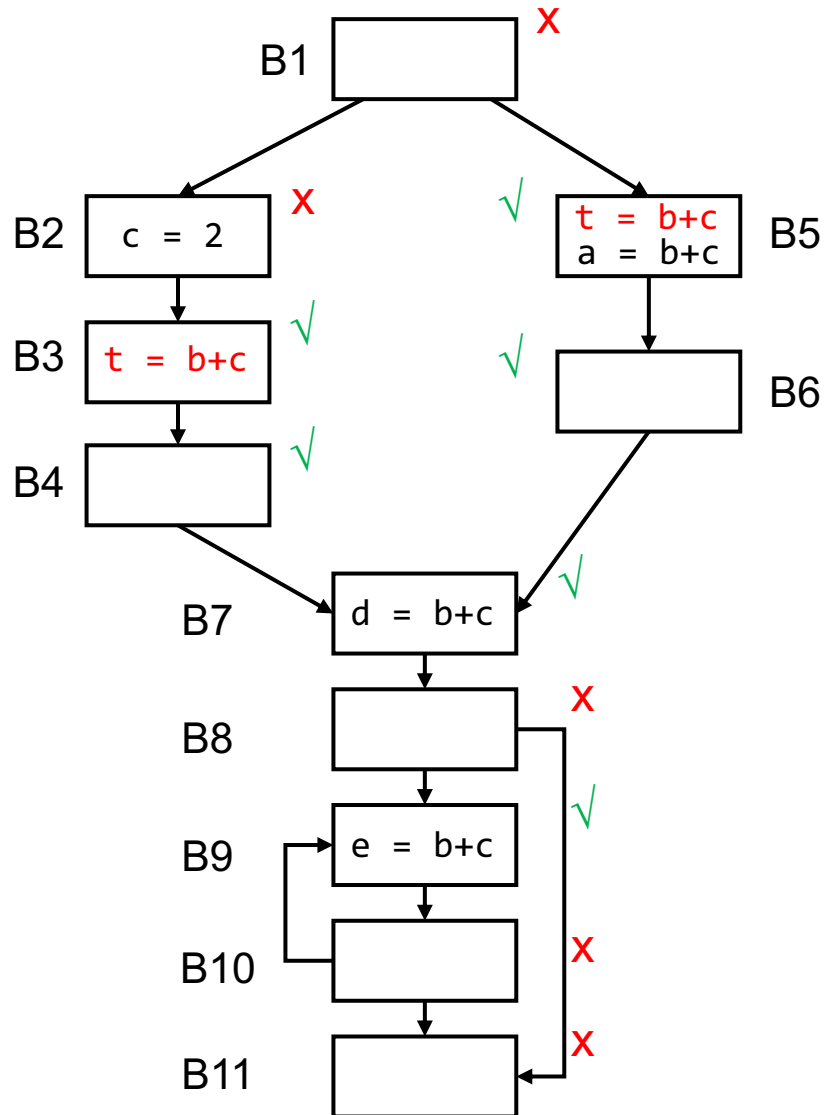
The Lazy Code Motion Problem

- All redundant computations of expressions that can be eliminated without code duplication are eliminated.
- The optimized program does not perform any computation that is not in the original program execution.
- Expressions are computed at the latest possible time.

The Lazy Code Motion Problem

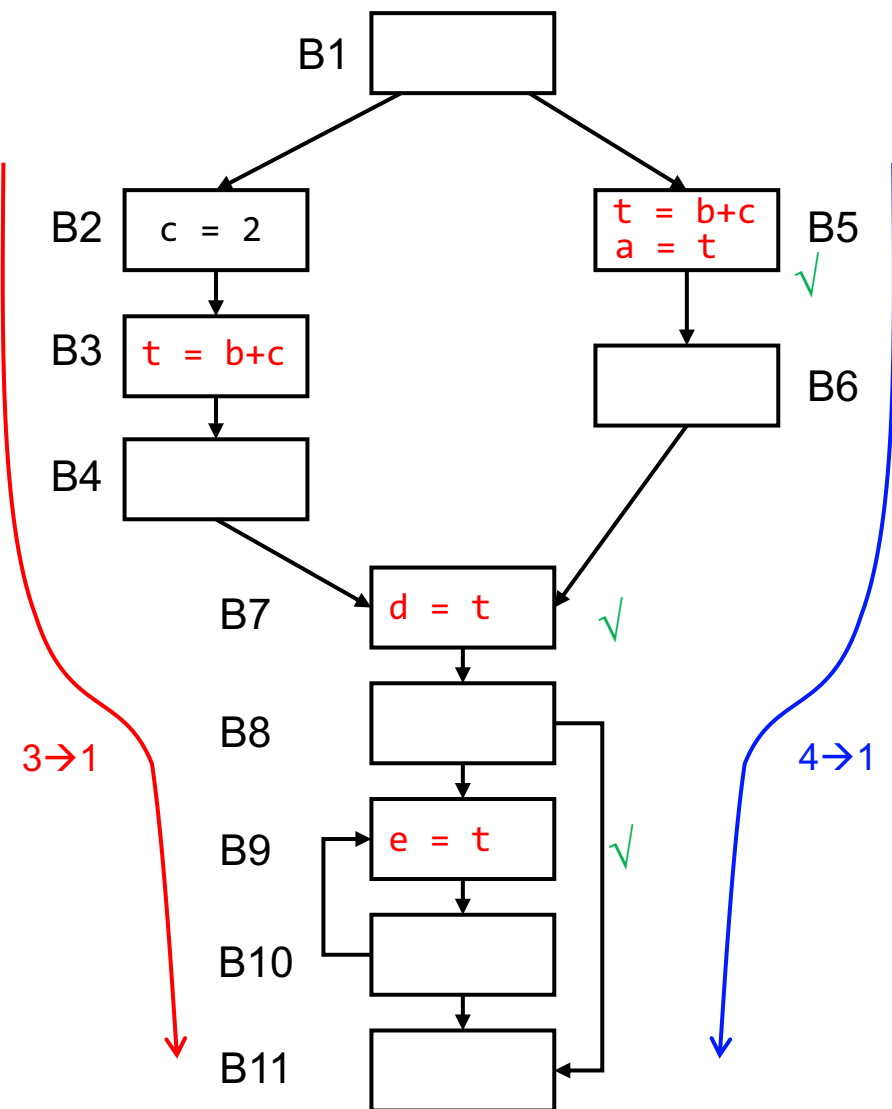
- **Step 1:** Anticipated Expressions (Backwards)
- **Step 2:** Available Expressions (Forwards)
- **Step 3:** Postponable Expressions (Forwards)
- **Step 4:** Used Expressions (Backwards)

Anticipated Expr (Backwards)



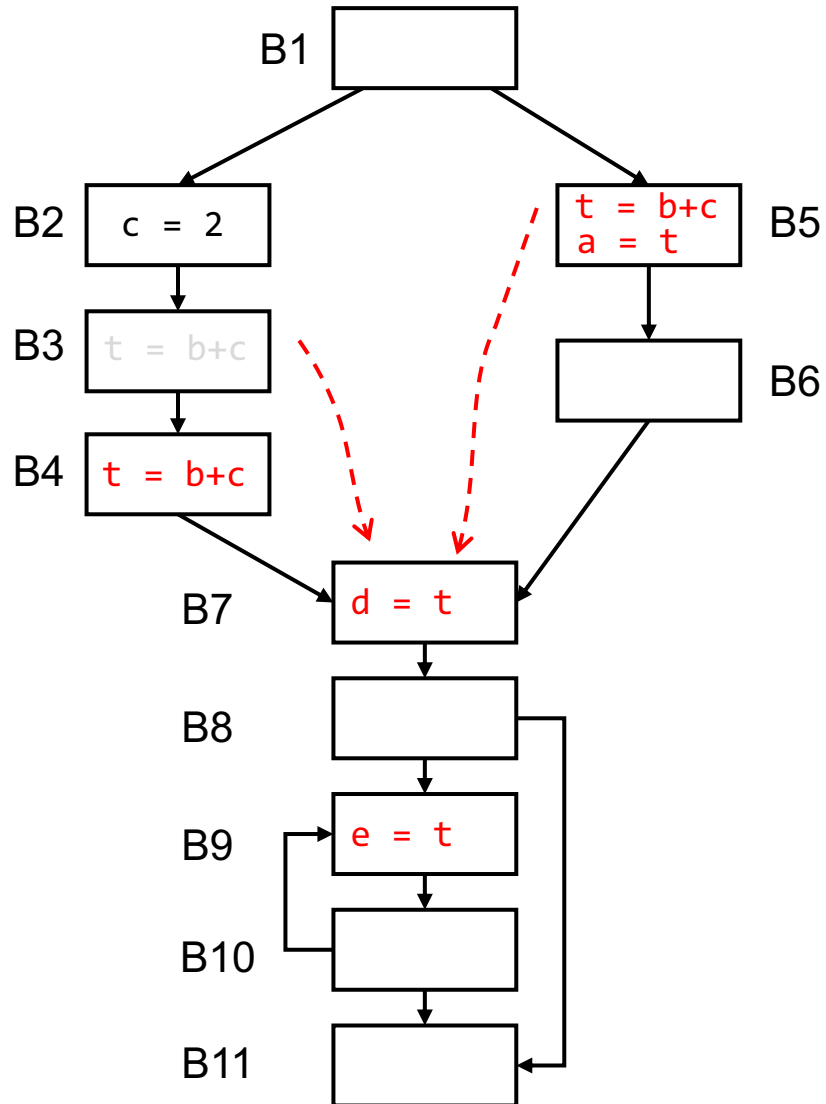
- An expression **b+c** is anticipated at a program point p if
 - all paths leading from point p eventually compute the expression **b+c** from the values of b and c that are available at that point

Available Expressions (Forwards)



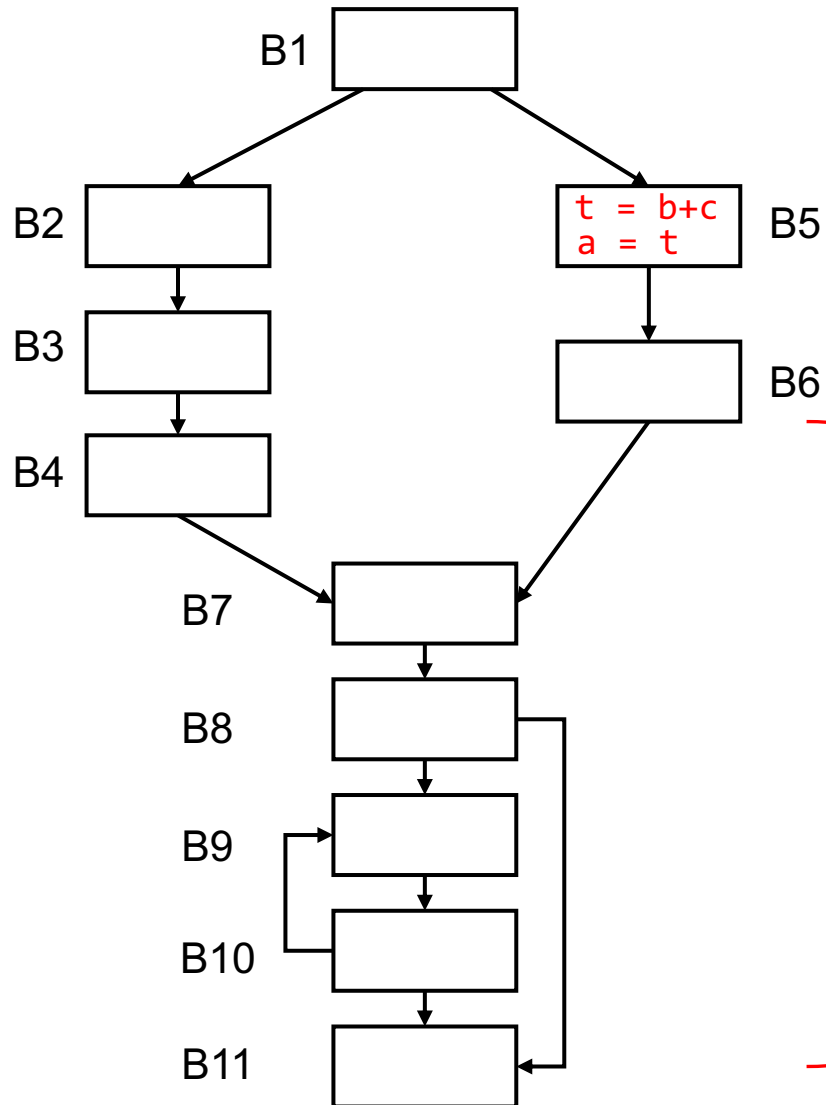
- Anticipated expr $\rightarrow$ Available expr
- An expression $b+c$ is available at program point p if it is anticipated along all paths reaching p .
- Replace $b+c$ with t .

Postponable Expr (Forwards)



- If an expression is far from its use, what may happen?
- Why is $t = b+c$ in B3 postponable?
- Why cannot $t = b+c$ in B3 be postponed to B7?

Used Expressions (Backwards)



- A simple backward data-flow pass to collect expressions used.
- Eliminate temporary variables that are used only once in the program.

Assume the variable t is never used from B6 to B11?

The Lazy Code Motion Problem

- **Step 1:** Anticipated Expressions (Backwards)
- **Step 2:** Available Expressions (Forwards)
- **Step 3:** Postponable Expressions (Forwards)
- **Step 4:** Used Expressions (Backwards)

Summary

	(a) Anticipated Expressions	(b) Available Expressions
Domain	Sets of expressions	Sets of expressions
Direction	Backwards	Forwards
Transfer function	$f_B(x) = e_use_B \cup (x - e_kill_B)$	$f_B(x) = (anticipated[B].in \cup x) - e_kill_B$
Boundary	$IN[EXIT] = \emptyset$	
Meet ($\wedge$)	$\cap$	
Equations	$IN[B] = f_B(OUT[B])$ $OUT[B] = \bigwedge_{S, succ(B)} IN[S]$	
Initialization	$IN[B] = U$	

	(c) Postponable Expressions	(d) Used Expressions
Domain	Sets of expressions	Sets of expressions
Direction	Forwards	Backwards
Transfer function	$f_B(x) = (earliest[B] \cup x) - e_use_B$	$f_B(x) = (e_use_B \cup x) - latest[B]$
Boundary	$OUT[ENTRY] = \emptyset$	$IN[EXIT] = \emptyset$
Meet ($\wedge$)	$\cap$	$\cup$
Equations	$OUT[B] = f_B(IN[B])$ $IN[B] = \bigwedge_{P, pred(B)} OUT[P]$	$IN[B] = f_B(OUT[B])$ $OUT[B] = \bigwedge_{S, succ(B)} IN[S]$
Initialization	$OUT[B] = U$	$IN[B] = \emptyset$

Refer to Chapter 9, the Dragon book!

PART III: Constant Propagation

Constant Propagation

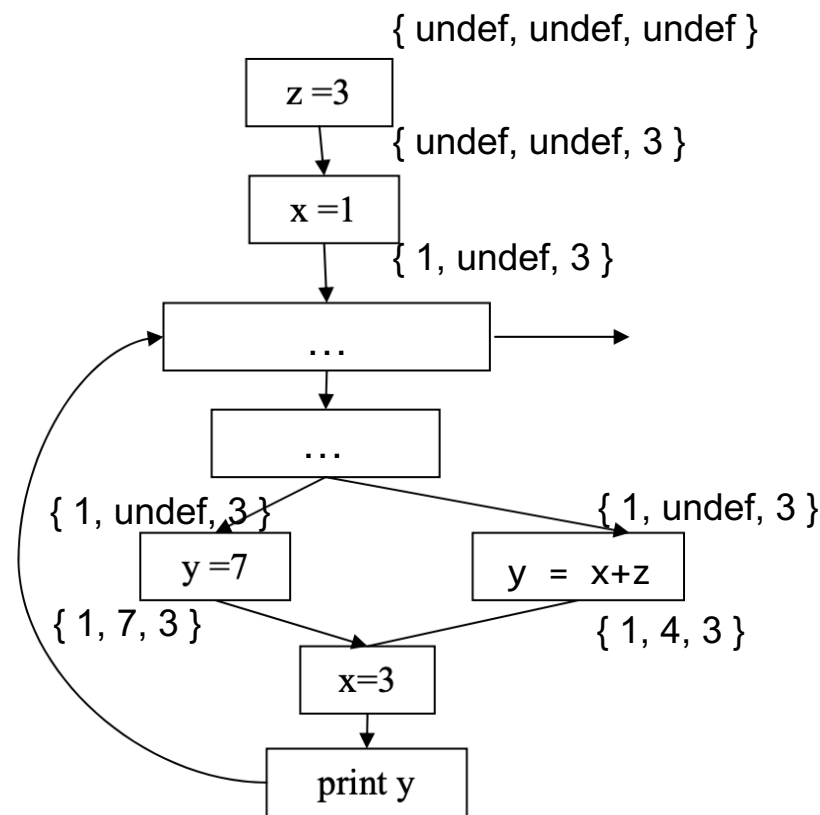
- Does a variable have a constant value at a program point?
- **Feature 1:** It has an unbounded set of possible data-flow values
- **Feature 2:** It is not distributive

Constant Propagation

- Each variable $v \rightarrow \text{UNDEF, Constant, NAC}$, denoted as $m(v)$

Transfer function of $x = y + z$

- $m(y) = \text{Constant}, m(z) = \text{Constant}$
- $\Rightarrow m(x) = \text{Constant}$
- either of $m(y)$ and $m(z)$ is UNDEF
- $\Rightarrow m(x) = \text{UNDEF}$
- otherwise
- $\Rightarrow m(x) = \text{NAC}$

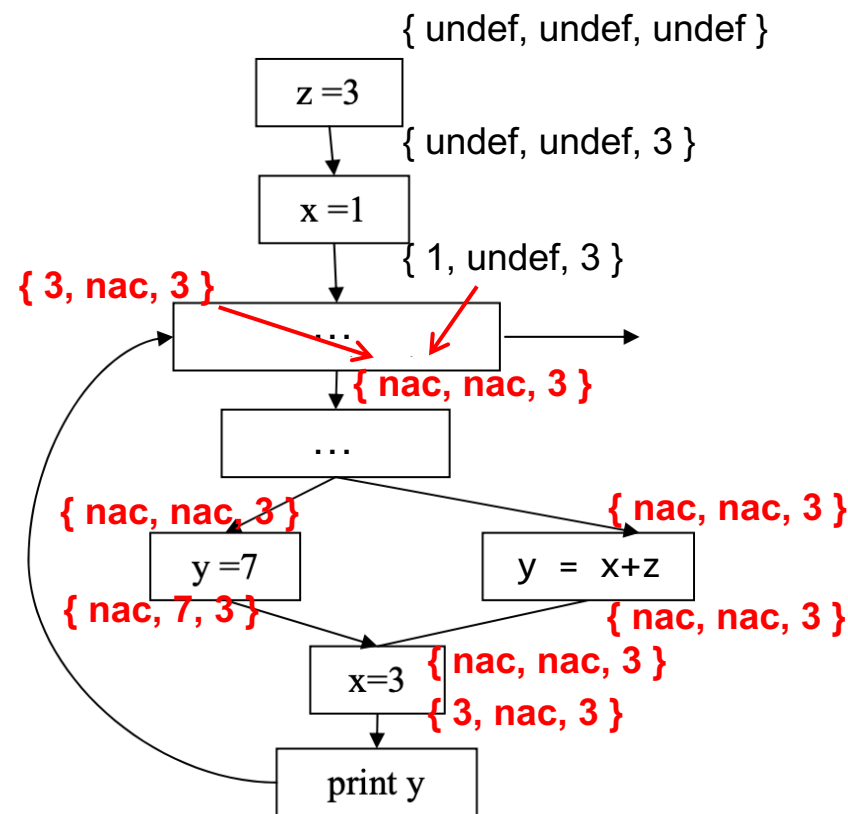


Constant Propagation

- Each variable $v \rightarrow \text{UNDEF, Constant, NAC}$, denoted as $m(v)$

Merge Function: $m(x) \wedge m(x')$

- $\text{UNDEF} \wedge \text{Constant} = \text{Constant}$
- $\text{NAC} \wedge \text{Constant} = \text{NAC}$
- $\text{Constant} \wedge \text{Constant} = \text{Constant}$
- $\text{Constant} \wedge \text{Constant}' = \text{NAC}$

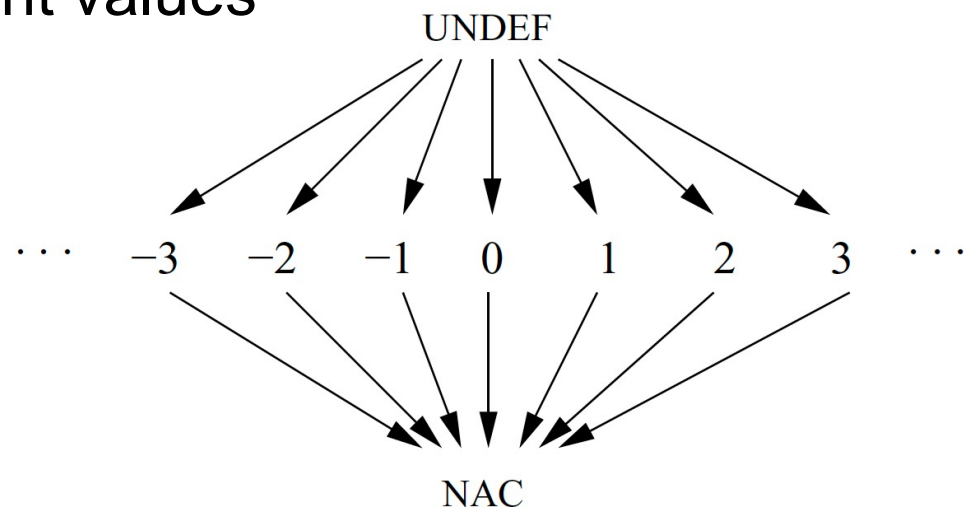


Constant Propagation

- Does a variable have a constant value at a program point?
- **Feature 1:** It has an unbounded set of possible data-flow values
- **Feature 2:** It is not distributive

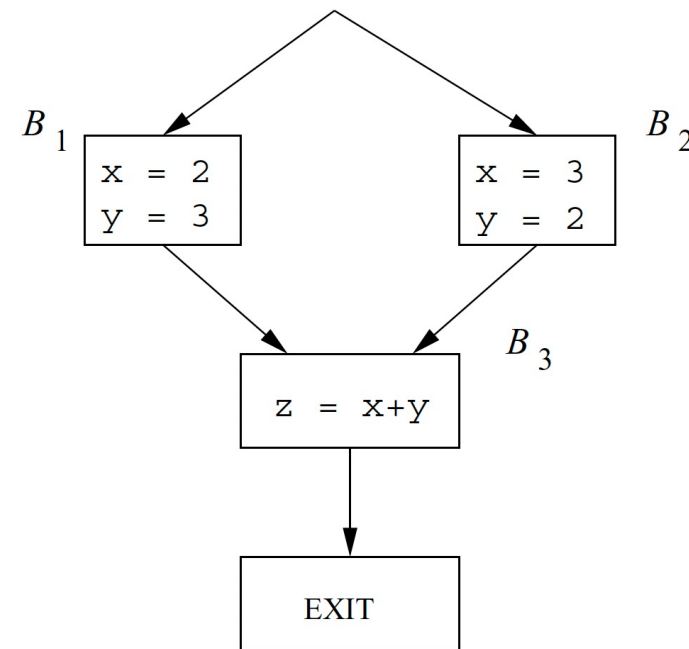
Monotonicity

- At each program point, $m(x)$ always starts with UNDEF
 - $\Rightarrow$ become constant after assignment
 - $\Rightarrow$ become NAC if assigned different values
- At each program point
 - # NAC always increases



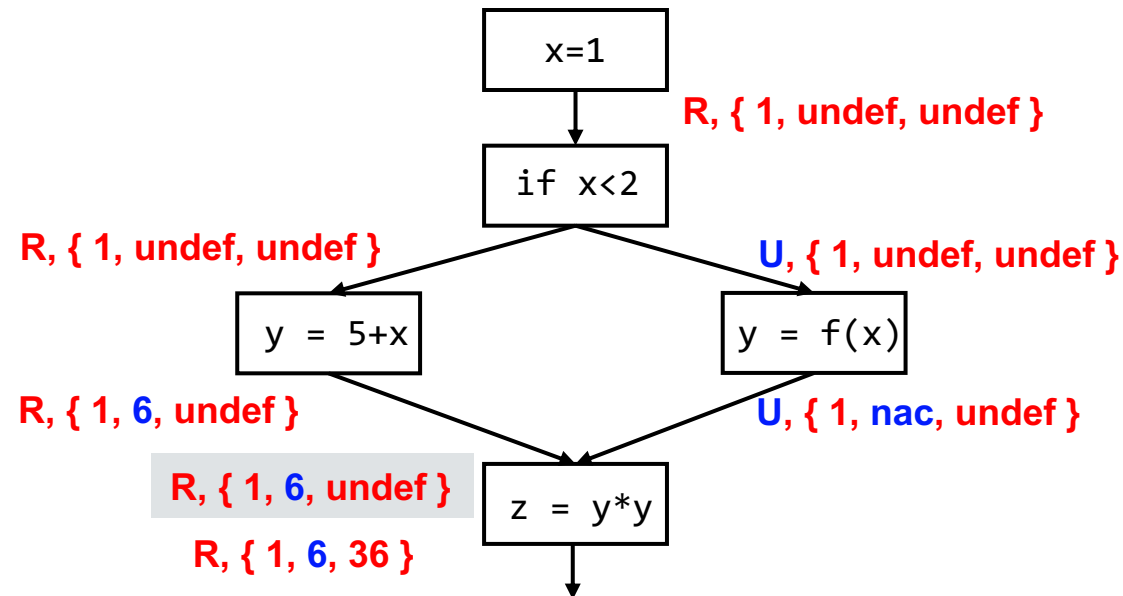
Non-distributivity

- $f(a \wedge b) \neq f(a) \wedge f(b) \Rightarrow$ Loss of precision in DFA!
- Meet-Then-Transfer $\neq$ Transfer-Then-Meet
- Meet-Then-Transfer
 - $m(x) = \text{NAC}, m(y) = \text{NAC} \Rightarrow m(z) = \text{NAC}$
- Transfer-Then-Meet
 - $m(z) = 5, m(z) = 5 \Rightarrow m(z) = 5$



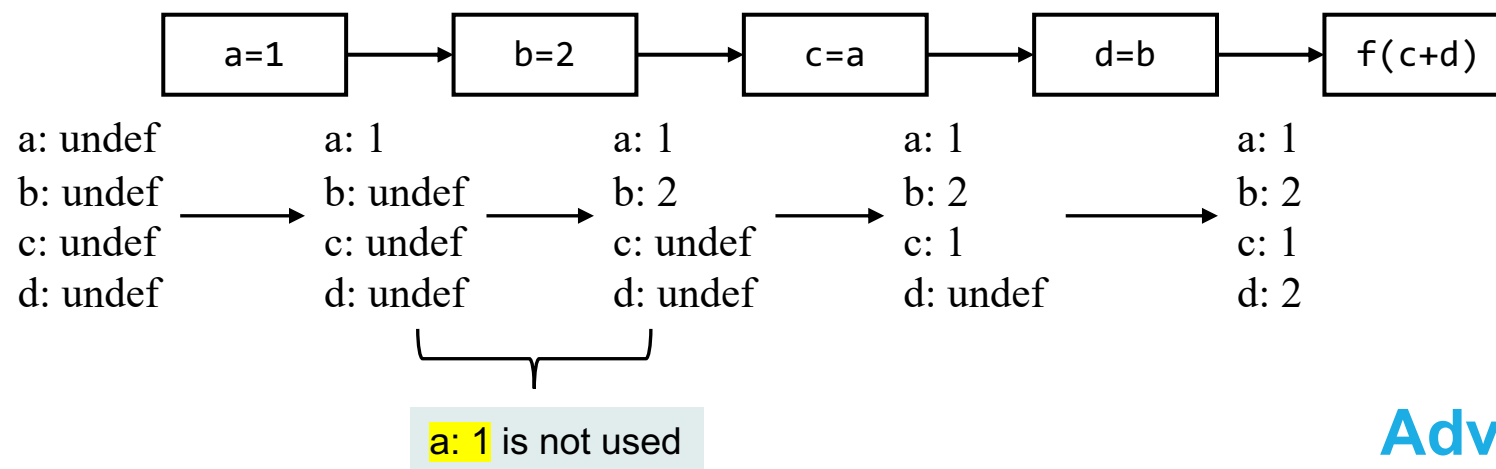
Conditional Const. Propagation

- Check the reachability of the code
- Check the branch condition => **Precision improvement!**

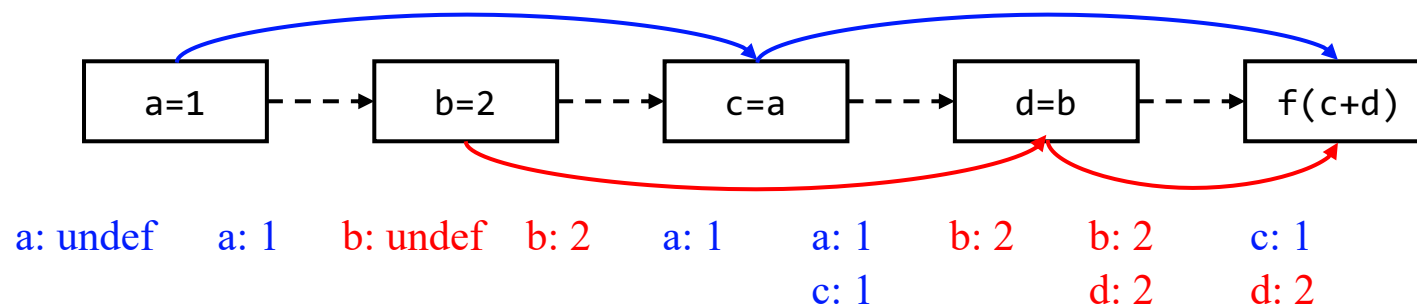


PART IV: Sparse DFA

Constant Propagation



Advantages?



Sparse DFA

- **Sparse Dataflow Analysis** **Relations to SSA?**
 - **Temporal sparsity:** Skip unnecessary control flows
 - **Space sparsity:** Preserve necessary dataflow facts at a program point

